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Faculty of Computer Science | Chair of Distributed and Networked Systems

# Advanced Computer Networks

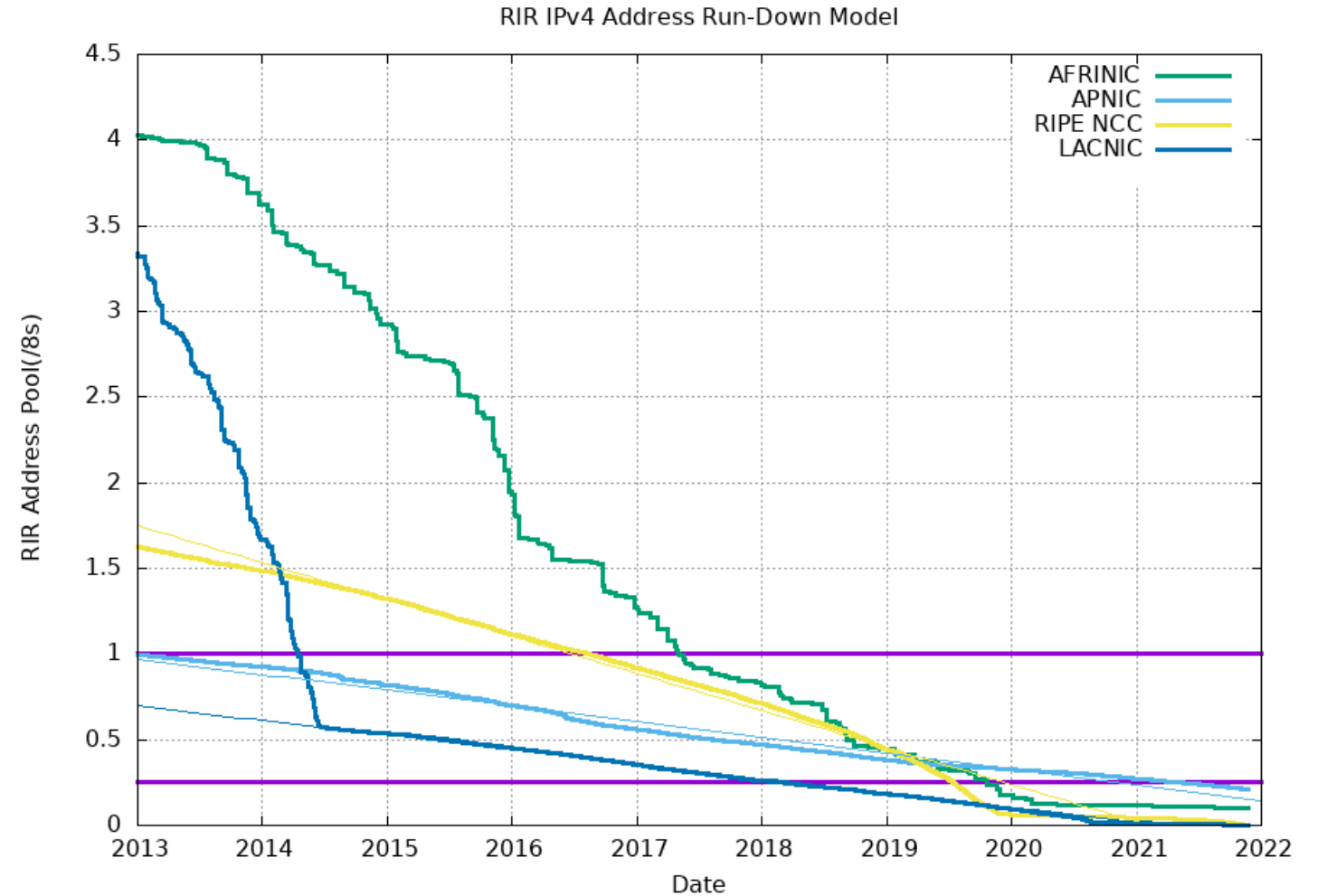
## IPv6 SRA + routing loops, and the Open DNS

# What happened last lecture?

IPv6

# “IPv4 Address Exhaustion” per RIR

Basically, **all address blocks have been assigned**. You may have a chance on the **waiting list**.



Source: <https://ipv4.potaroo.net/>

# End-To-End Arguments in System Design

J. H. SALTZER, D. P. REED, and D. D. CLARK

Massachusetts Institute of Technology Laboratory for Computer Science

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This paper presents a design principle that helps guide placement of functions among the modules of a distributed computer system. The principle, called the end-to-end argument, suggests that functions

**NAT is not a solution for a proper Internet.**

*architectures*; C.2.2 [**Computer-Communication Networks**]: Network Protocols—*protocol architecture*; C.2.4 [**Computer-Communication Networks**]: Distributed Systems; D.4.7 [**Operating Systems**]: Organization and Design—*distributed systems*

General Terms: Design

Additional Key Words and Phrases: Data communication, protocol design, design principles

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[ACM Transactions on Computer Systems, Nov. 1984, <https://dl.acm.org/doi/10.1145/357401.357402>]

# Notation of IPv6 addresses

$2^{128}$  Bit

X 16 Bit Hex

X : X : X : X : X : X : X : X : X : X

Example

2001 : 0DB8 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001

# IPv6 Innovations: Simpler **addressing** and **routing**

Elimination of address bottleneck: 128 Bit addresses

Several addresses per interface are common

# IPv6 Innovations: Simpler protocol architecture

Slim, fixed header for fast processing

Optional extension headers

Format framework for header classes

No header checksum (already done by layer 2 and 4)

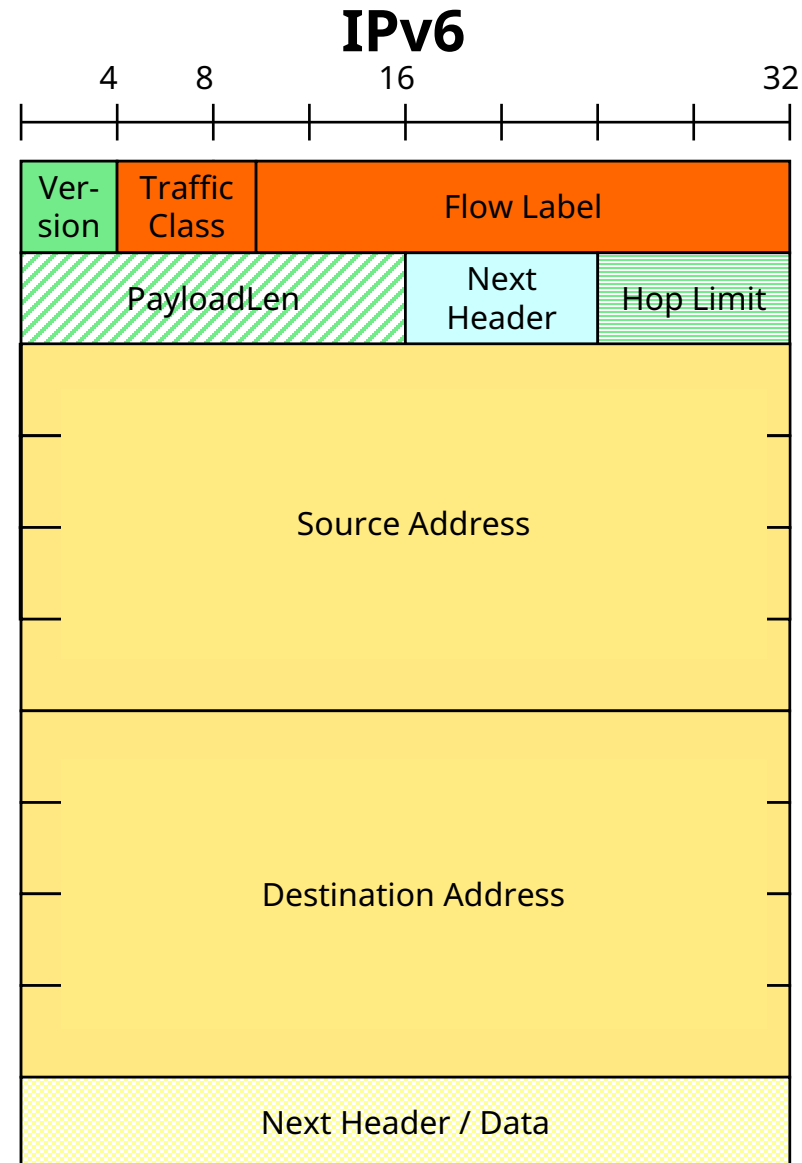
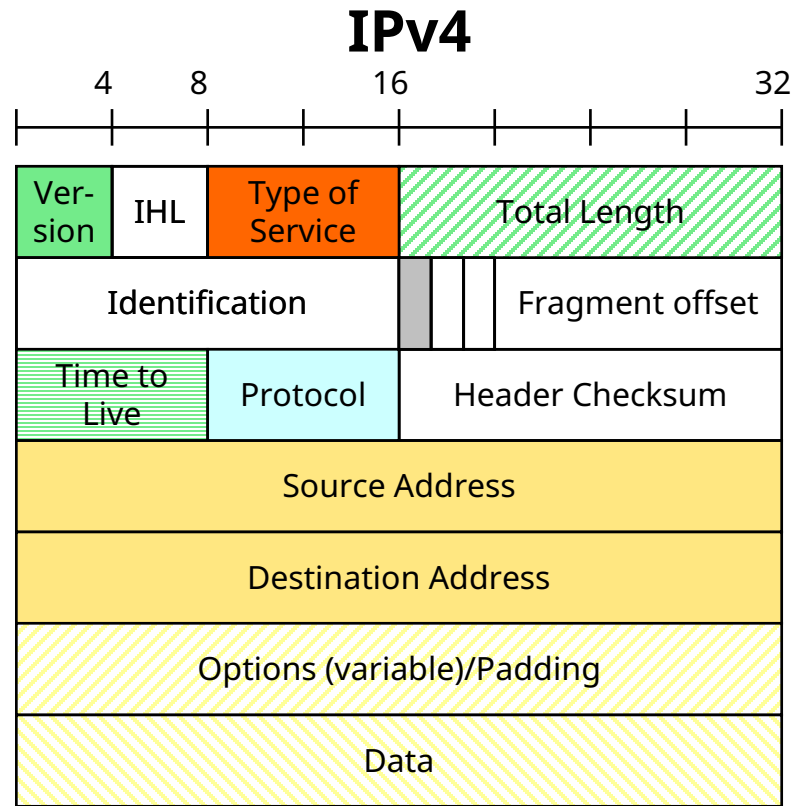
No fragmentation in routers

# IPv6 Innovations: Simpler **administration**

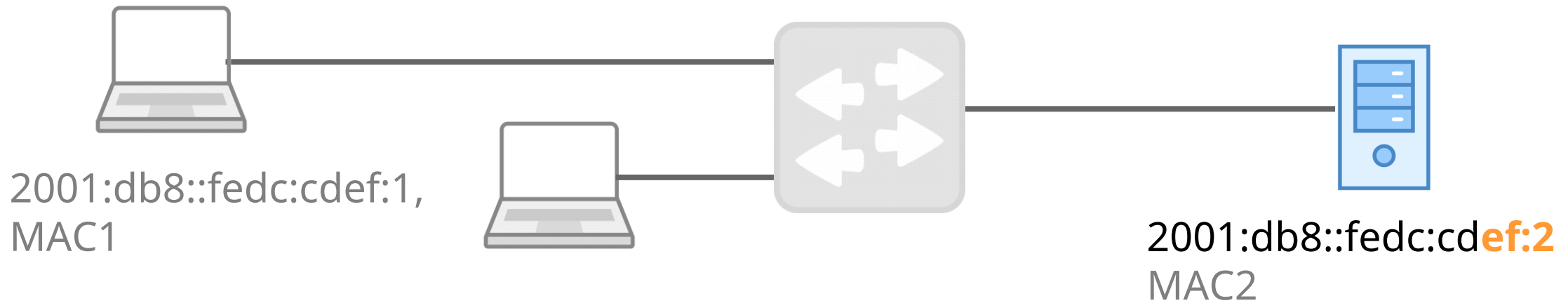
Stateless Address Autoconfiguration (SLAAC) of interfaces without DHCPv6

Renumbering via prefix change

# Packet Formats: IPv6 vs. IPv4



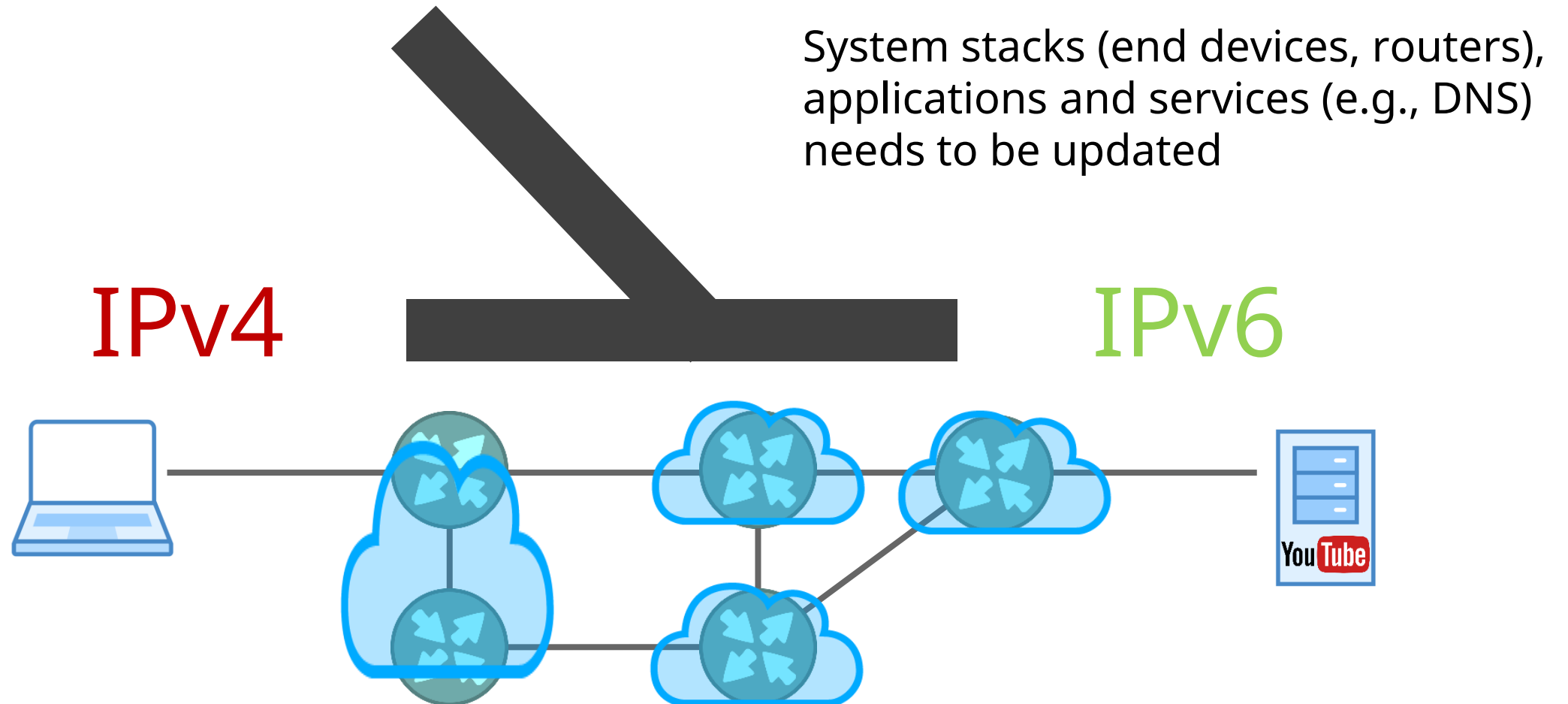
# IPv6 Neighbor Discovery Protocol. ND uses Multicast



Neighbor Solicitation [ IP DST: **ff02::1:ff00:ef:2** ]

Solicited Node Multicast Address: ff02::1:ff**00:0000**/104  
Remaining 24 bits encode the last 24 bits of the requested IPv6 address.

# We cannot replace IPv4 by night



# Transition from IPv4 to IPv6

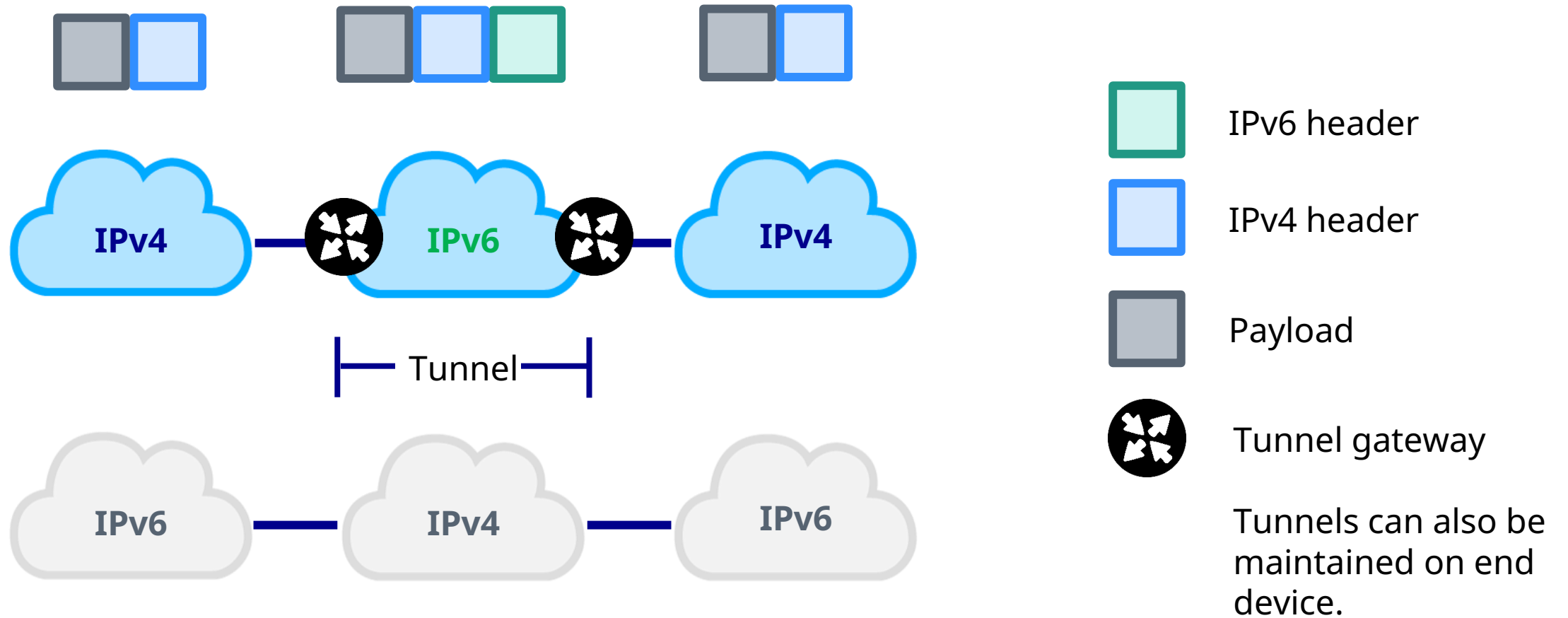
## Porting

Make applications IPv6-ready by using the new sockets API, new address structures

## Migration

Incremental change from v4 to v6; temporarily use both protocols in parallel by deploying dual stack, tunnels, protocol translators

# Dual stack does not help in all scenarios. Migration by tunneling.



# Challenges when tunneling

Learning the tunnel end point

Which v6 (or v4) tunnel endpoint is responsible for which v4 (or v6) IP address?

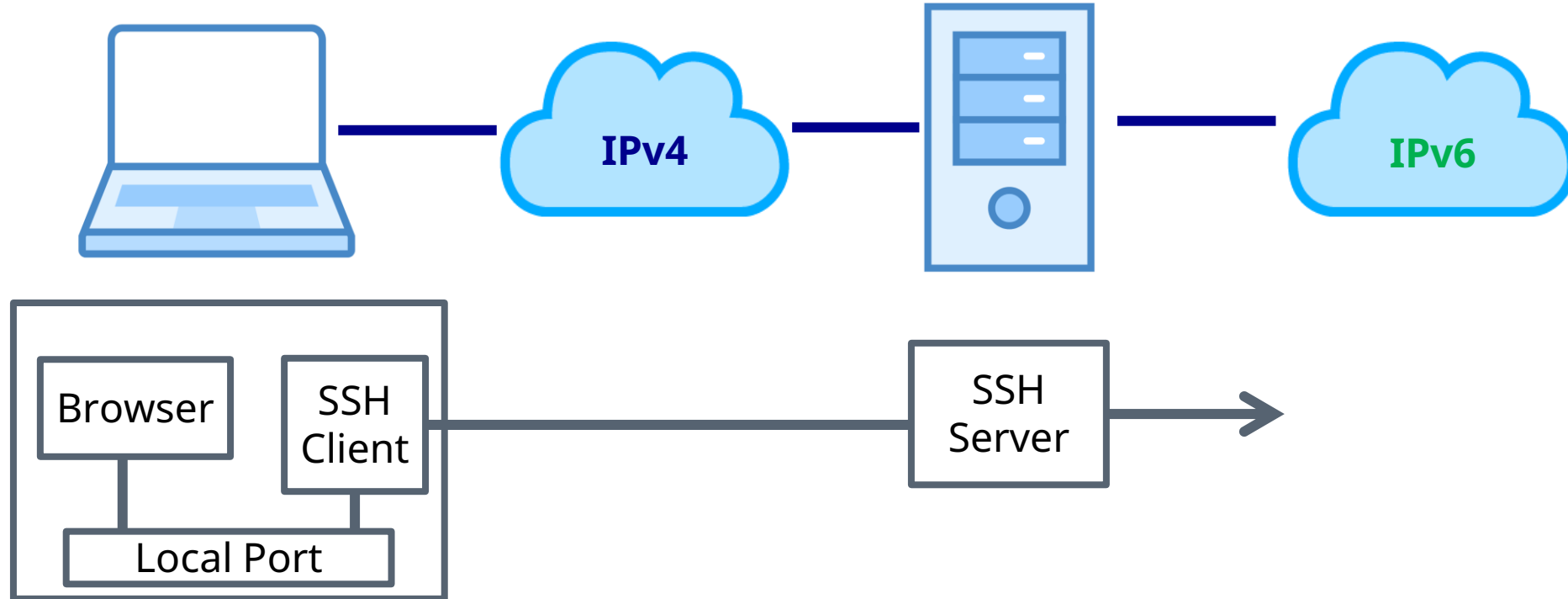
More overhead because of two headers

Additional headers increase packets sizes

Increased packet sizes may lead to fragmentation

Packet fragmentation introduces additional challenges

# SSH IPv6 Tunnel (port forwarding) + SOCKS Proxy



SSH client opens TCP port locally and connects to the SSH server  
Browser connects to local TCP using SOCKS Proxy protocol

# What happens this lecture?

IPv6 SRA + routing loops, and the Open DNS

IPv6 SRA + routing loops

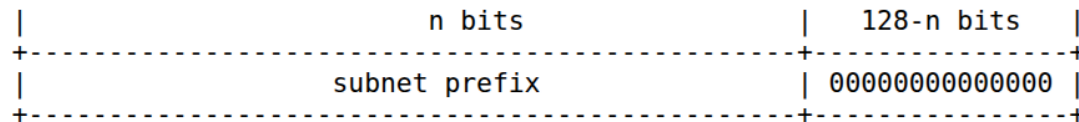
# Subnet-Router Anycast (SRA) addresses

A specific IPv6 address to find the closest router of a subnet

## RFC 1884 (IPv6 addressing architecture):

### 2.5.1 Required Anycast Address

The Subnet-Router anycast address is predefined. It's format is as follows:



**Routers with interfaces** to the corresponding subnet are **required** to reply!

# Subnet-Router Anycast (SRA) addresses

A specific IPv6 address to find the closest router of a subnet

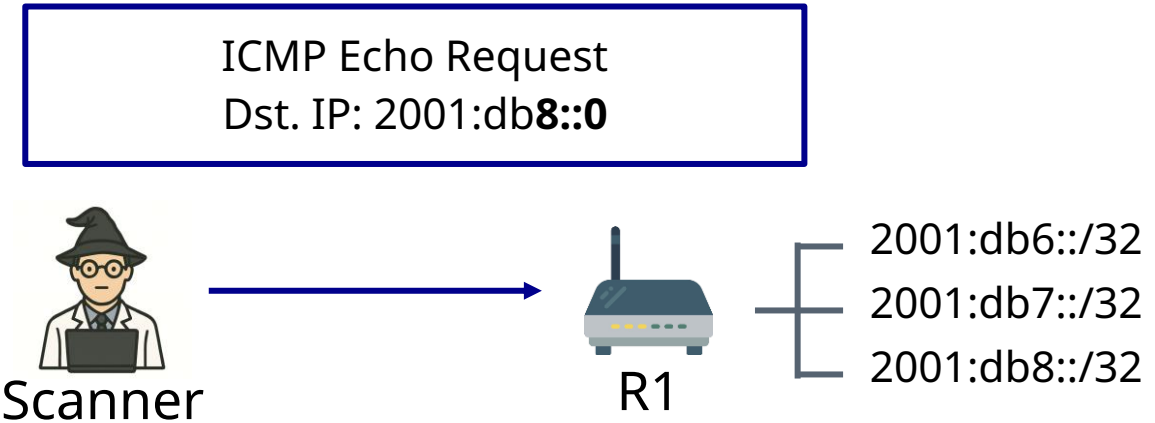
## RFC 1884 (IPv6 addressing architecture):

### 2.5.1 Required Anycast Address

The Subnet-Router anycast address is predefined. It's format is as follows:

|   |               |   |                  |   |
|---|---------------|---|------------------|---|
|   | n bits        |   | 128-n bits       |   |
| + | -----         | + | -----            | + |
|   | subnet prefix |   | 0000000000000000 |   |
| + | -----         | + | -----            | + |

**Routers with interfaces** to the corresponding subnet are **required** to reply!



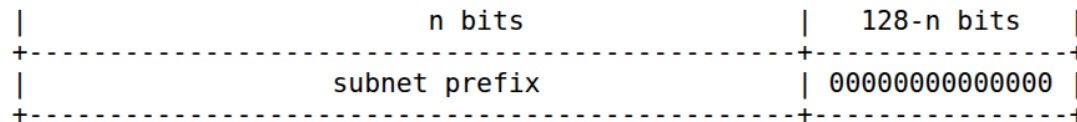
# Subnet-Router Anycast (SRA) probing

A specific IPv6 address to find the closest router of a subnet

## RFC 1884 (IPv6 addressing architecture):

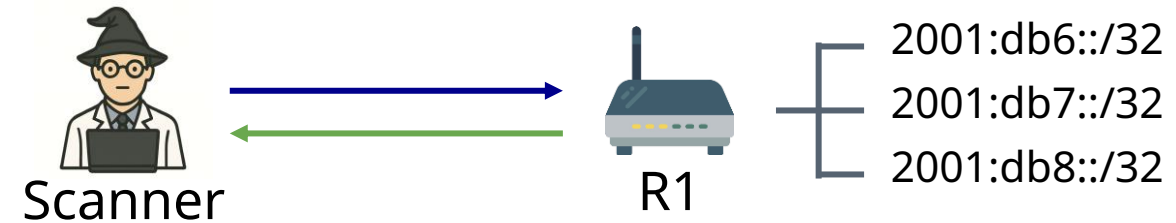
### 2.5.1 Required Anycast Address

The Subnet-Router anycast address is predefined. It's format is as follows:



**Routers with interfaces** to the corresponding subnet are **required** to reply!

ICMP Echo Request  
Dst. IP: 2001:db8::0

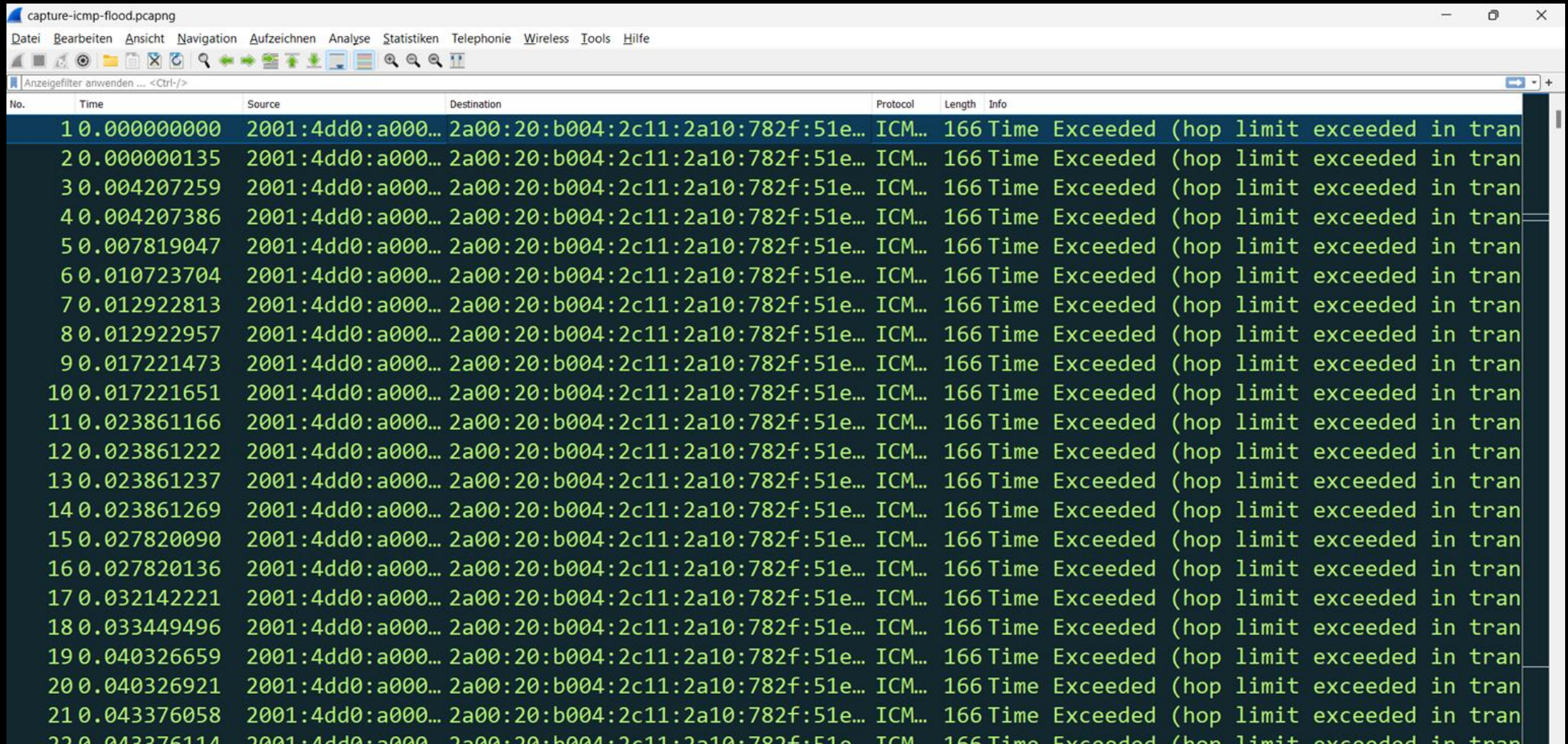


ICMP Echo Reply  
Src. IP: R1

**Helpful in case of IPv6 scanning**  
since it reduces ICMPv6 error messages and thus rate limiting [see Internet measurements].

```
ping -c 1 <destination IP address>
```

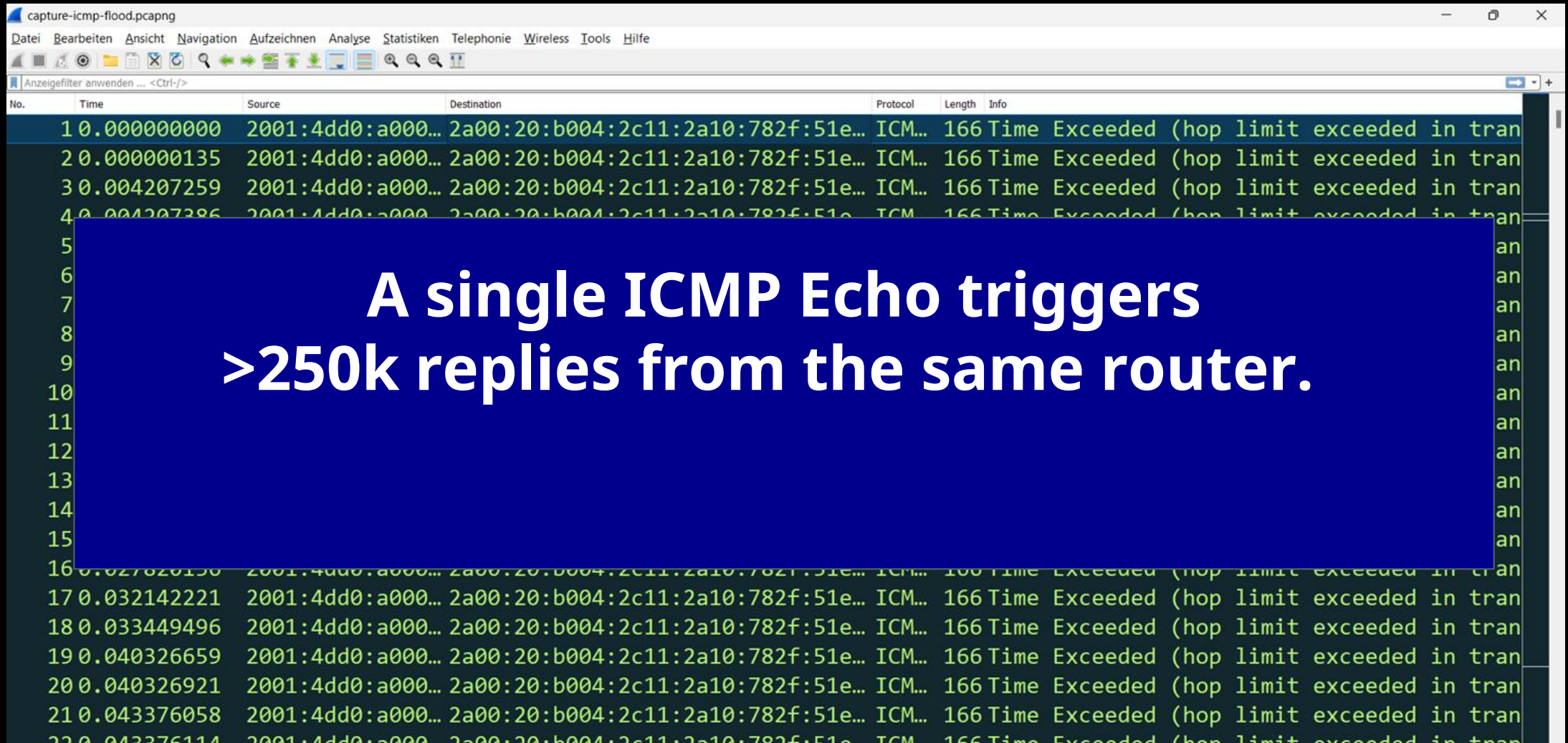
ping -c 1 <destination IP address>



The image shows a Wireshark packet capture window titled 'capture-icmp-flood.pcapng'. The interface includes a menu bar (Datei, Bearbeiten, Ansicht, Navigation, Aufzeichnen, Analyse, Statistiken, Telefonie, Wireless, Tools, Hilfe) and a toolbar with various icons. Below the toolbar is a filter bar with the text 'Anzeigefilter anwenden ... <Ctrl-/>'. The main display area shows a list of 22 captured packets. Each packet is an ICMP Echo (ping) request from source 2001:4dd0:a000::1 to destination 2a00:20:b004:2c11:2a10:782f:51e::1. All packets have a length of 166 bytes and the information field contains the message 'Exceeded (hop limit exceeded in transit)'. The time interval between packets is approximately 0.004 seconds.

| No. | Time        | Source            | Destination                        | Protocol | Length | Info                                          |
|-----|-------------|-------------------|------------------------------------|----------|--------|-----------------------------------------------|
| 1   | 0.000000000 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 2   | 0.000000135 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 3   | 0.004207259 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 4   | 0.004207386 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 5   | 0.007819047 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 6   | 0.010723704 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 7   | 0.012922813 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 8   | 0.012922957 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 9   | 0.017221473 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 10  | 0.017221651 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 11  | 0.023861166 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 12  | 0.023861222 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 13  | 0.023861237 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 14  | 0.023861269 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 15  | 0.027820090 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 16  | 0.027820136 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 17  | 0.032142221 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 18  | 0.033449496 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 19  | 0.040326659 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 20  | 0.040326921 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 21  | 0.043376058 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |
| 22  | 0.043376114 | 2001:4dd0:a000::1 | 2a00:20:b004:2c11:2a10:782f:51e::1 | ICMP...  | 166    | Time Exceeded (hop limit exceeded in transit) |

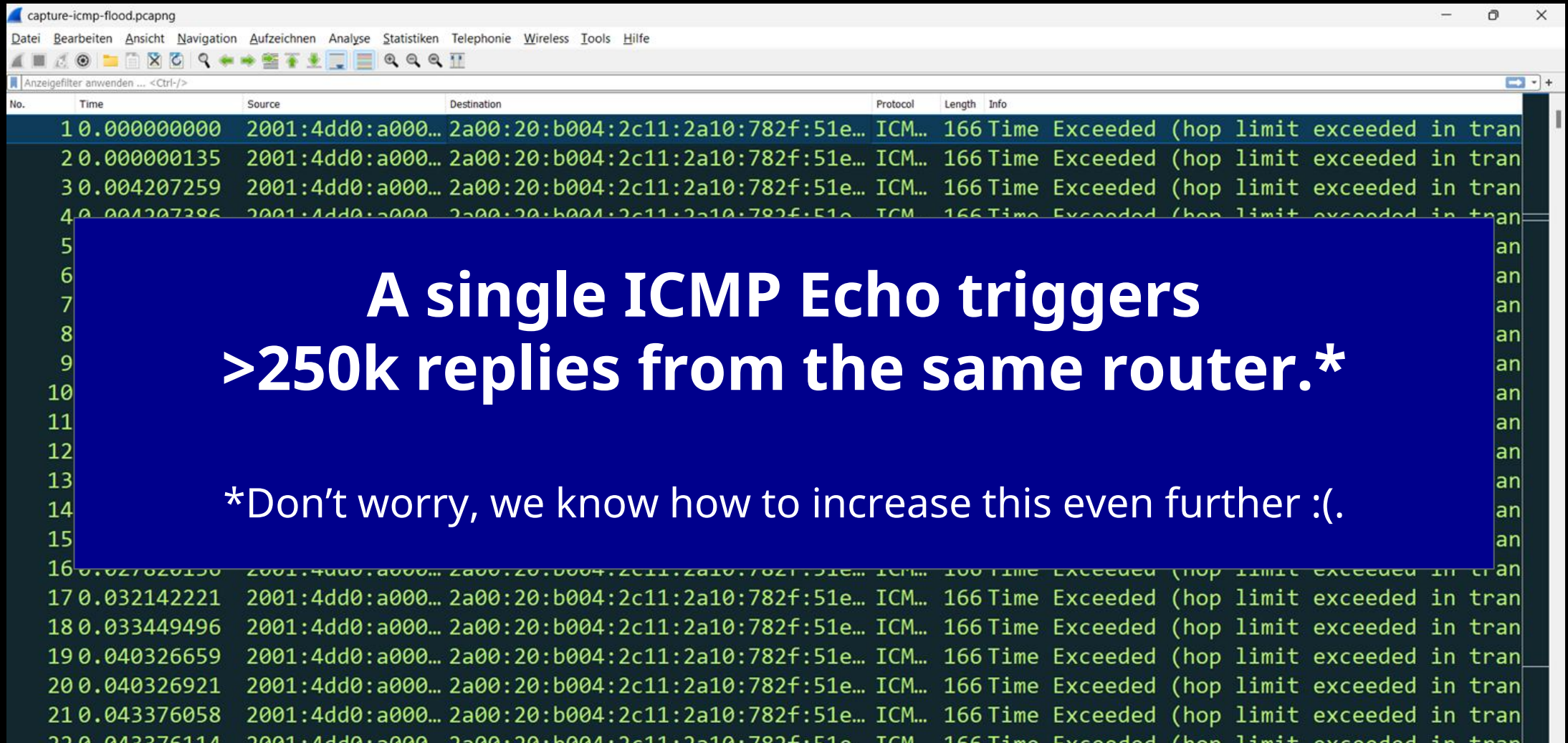
```
ping -c 1 <destination IP address>
```



The image shows a Wireshark packet capture window titled 'capture-icmp-flood.pcapng'. The interface includes a menu bar (Datei, Bearbeiten, Ansicht, Navigation, Aufzeichnen, Analyse, Statistiken, Telefonie, Wireless, Tools, Hilfe) and a toolbar. Below the toolbar is a filter bar with the text 'Anzeigefilter anwenden ... <Ctrl-/>'. The main display area shows a list of network packets. The first four packets are visible, all being ICMP Echo (ping) requests. A large blue text box is overlaid on the packet list, containing the text: 'A single ICMP Echo triggers >250k replies from the same router.' The packet list columns are: No., Time, Source, Destination, Protocol, Length, and Info. The packets show a source IP of 2001:4dd0:a000... and a destination IP of 2a00:20:b004:2c11:2a10:782f:51e... The protocol is ICMP, and the length is 166 bytes. The info field for the first four packets indicates 'Time Exceeded (hop limit exceeded in tran...'.

| No. | Time        | Source            | Destination                        | Protocol | Length | Info                                      |
|-----|-------------|-------------------|------------------------------------|----------|--------|-------------------------------------------|
| 1   | 0.000000000 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran |
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| 3   | 0.004207259 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran |
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| 5   |             |                   |                                    |          |        | an                                        |
| 6   |             |                   |                                    |          |        | an                                        |
| 7   |             |                   |                                    |          |        | an                                        |
| 8   |             |                   |                                    |          |        | an                                        |
| 9   |             |                   |                                    |          |        | an                                        |
| 10  |             |                   |                                    |          |        | an                                        |
| 11  |             |                   |                                    |          |        | an                                        |
| 12  |             |                   |                                    |          |        | an                                        |
| 13  |             |                   |                                    |          |        | an                                        |
| 14  |             |                   |                                    |          |        | an                                        |
| 15  |             |                   |                                    |          |        | an                                        |
| 16  | 0.027820150 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran |
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| 18  | 0.033449496 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran |
| 19  | 0.040326659 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran |
| 20  | 0.040326921 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran |
| 21  | 0.043376058 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran |
| 22  | 0.043376114 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran |

```
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```



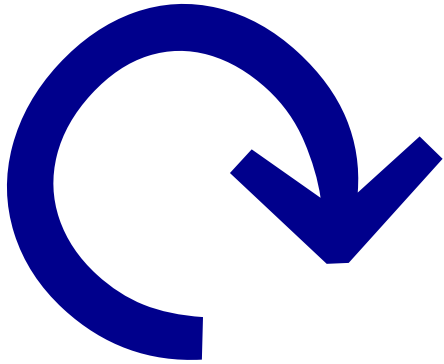
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| No. | Time        | Source            | Destination                        | Protocol | Length | Info                                         |
|-----|-------------|-------------------|------------------------------------|----------|--------|----------------------------------------------|
| 1   | 0.000000000 | 2001:4dd0:a000... | 2a00:20:b004:2c11:2a10:782f:51e... | ICMP...  | 166    | Time Exceeded (hop limit exceeded in tran... |
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| 6   |             |                   |                                    |          |        | an                                           |
| 7   |             |                   |                                    |          |        | an                                           |
| 8   |             |                   |                                    |          |        | an                                           |
| 9   |             |                   |                                    |          |        | an                                           |
| 10  |             |                   |                                    |          |        | an                                           |
| 11  |             |                   |                                    |          |        | an                                           |
| 12  |             |                   |                                    |          |        | an                                           |
| 13  |             |                   |                                    |          |        | an                                           |
| 14  |             |                   |                                    |          |        | an                                           |
| 15  |             |                   |                                    |          |        | an                                           |
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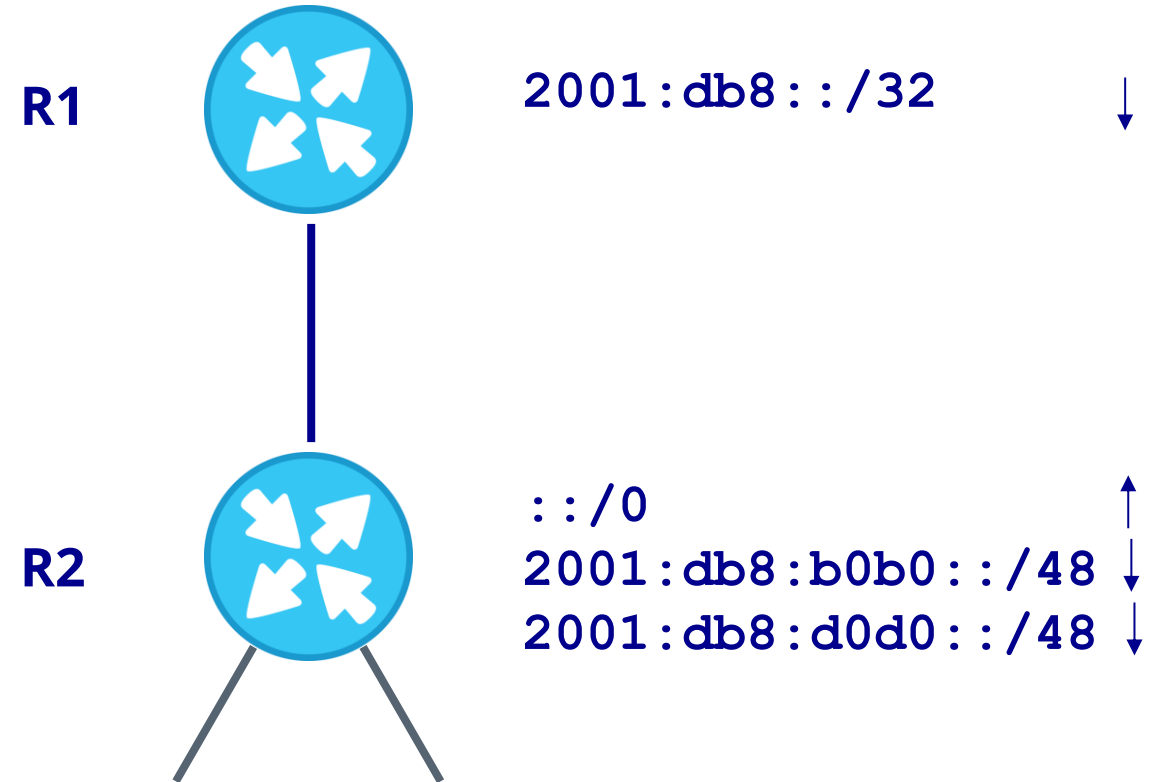
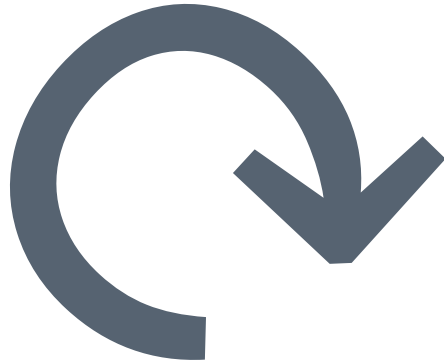
# Why?



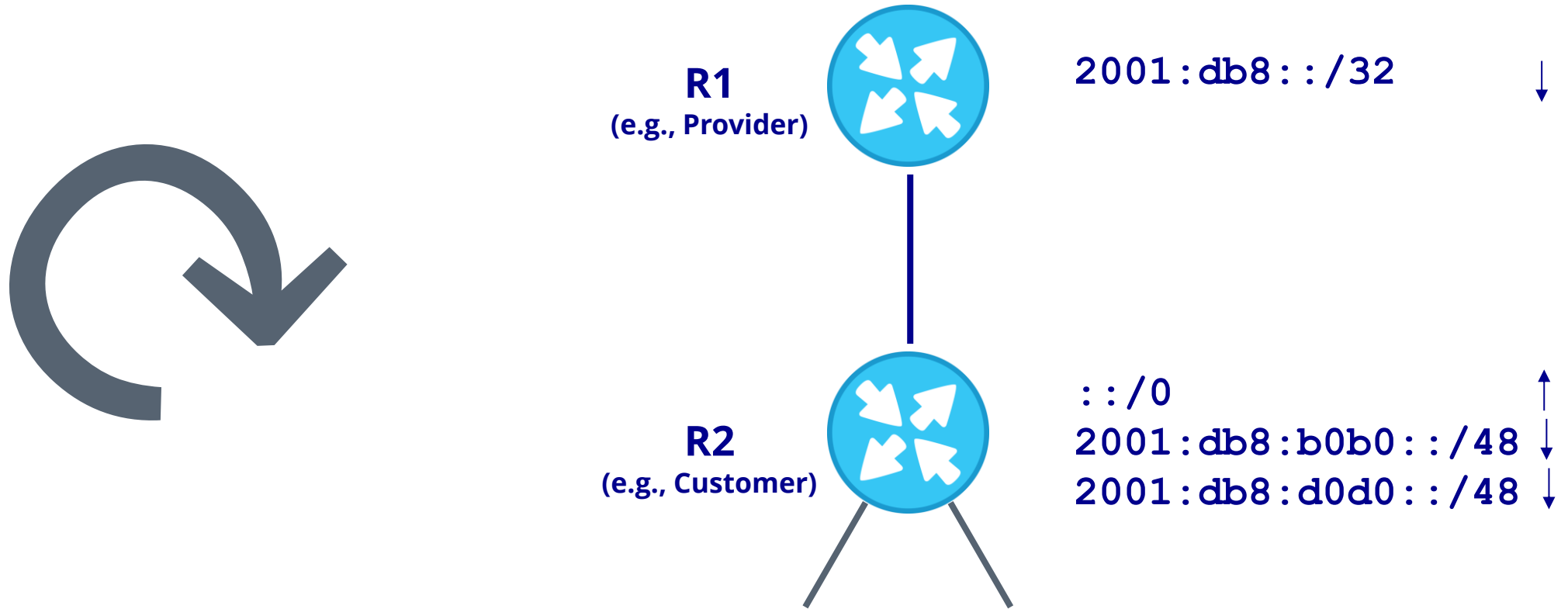
# What is a routing loop?



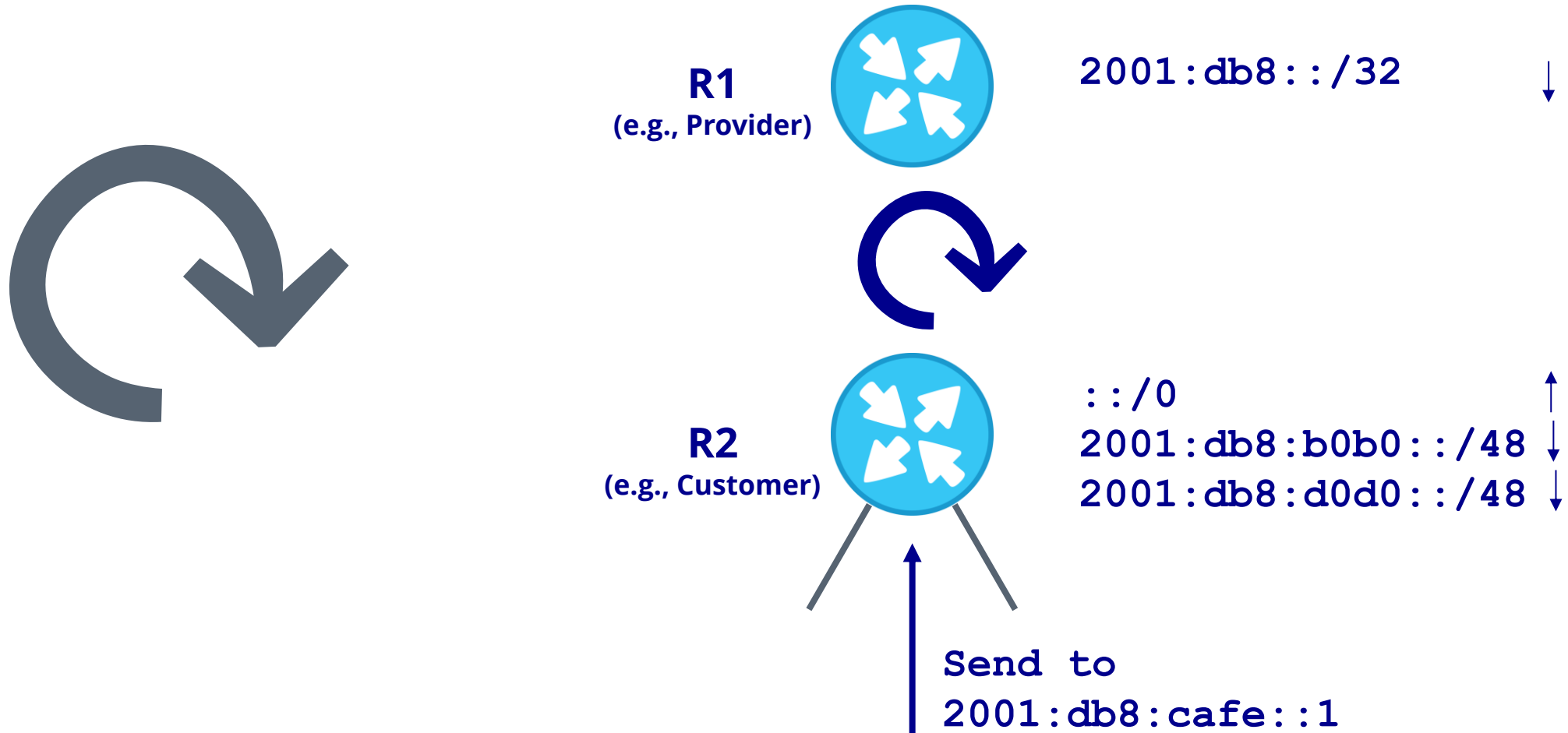
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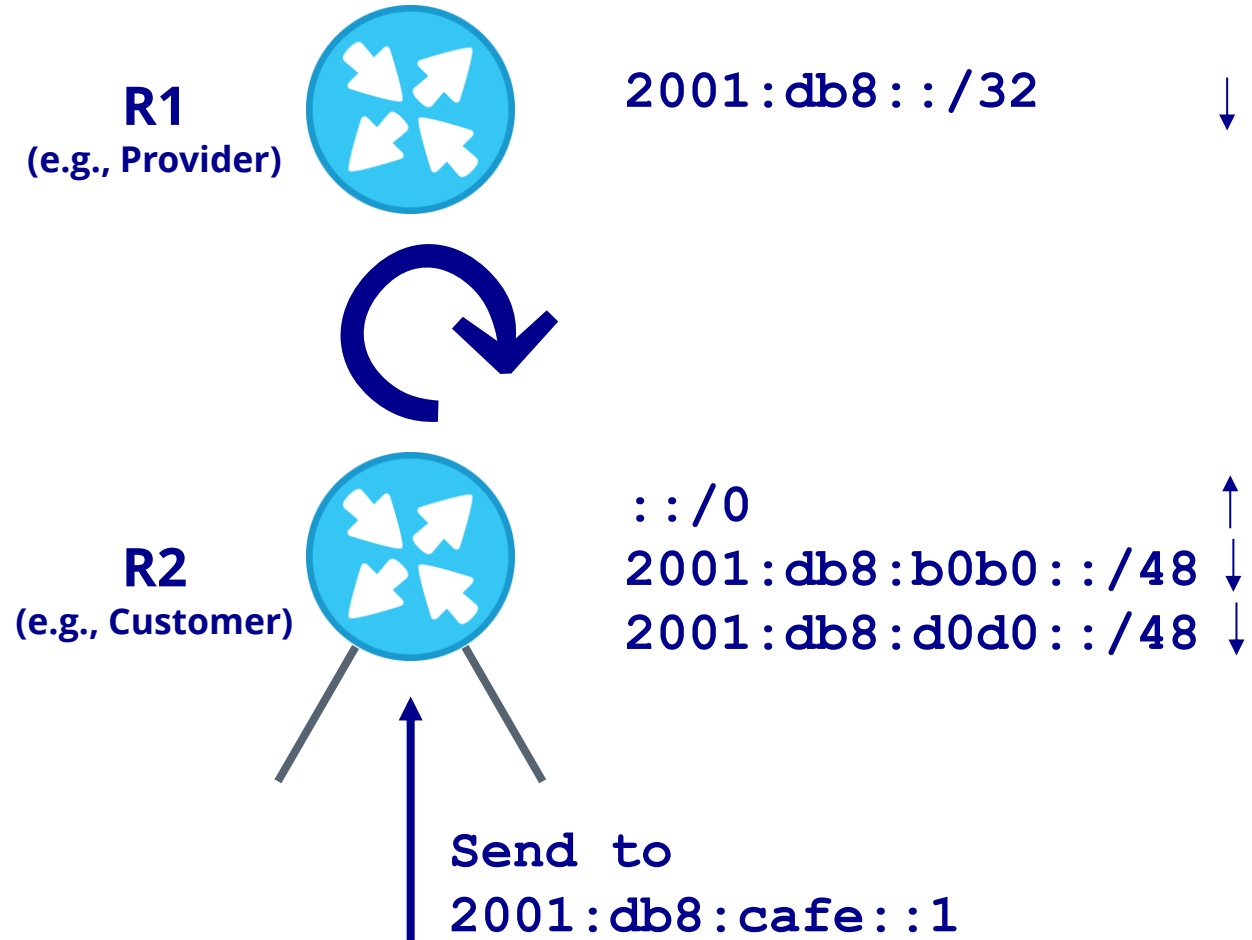
# What is a routing loop?



# Common deployment? Yes!

Those deployments **easily occur when providers assign PA address space to customers.**

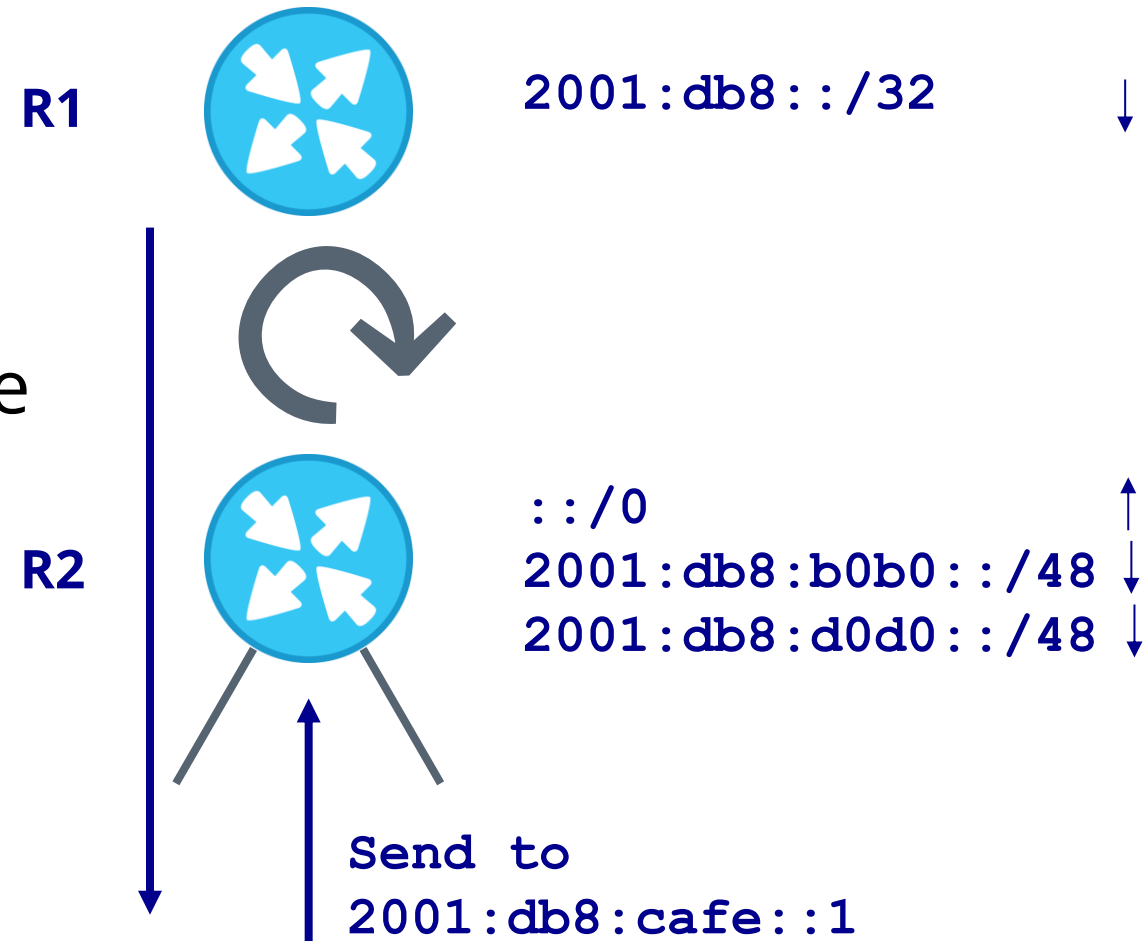
... and there is one more problem.



# A loop ultimately leads to an ICMP TTL Exceeded

# ICMP TTL Exceeded

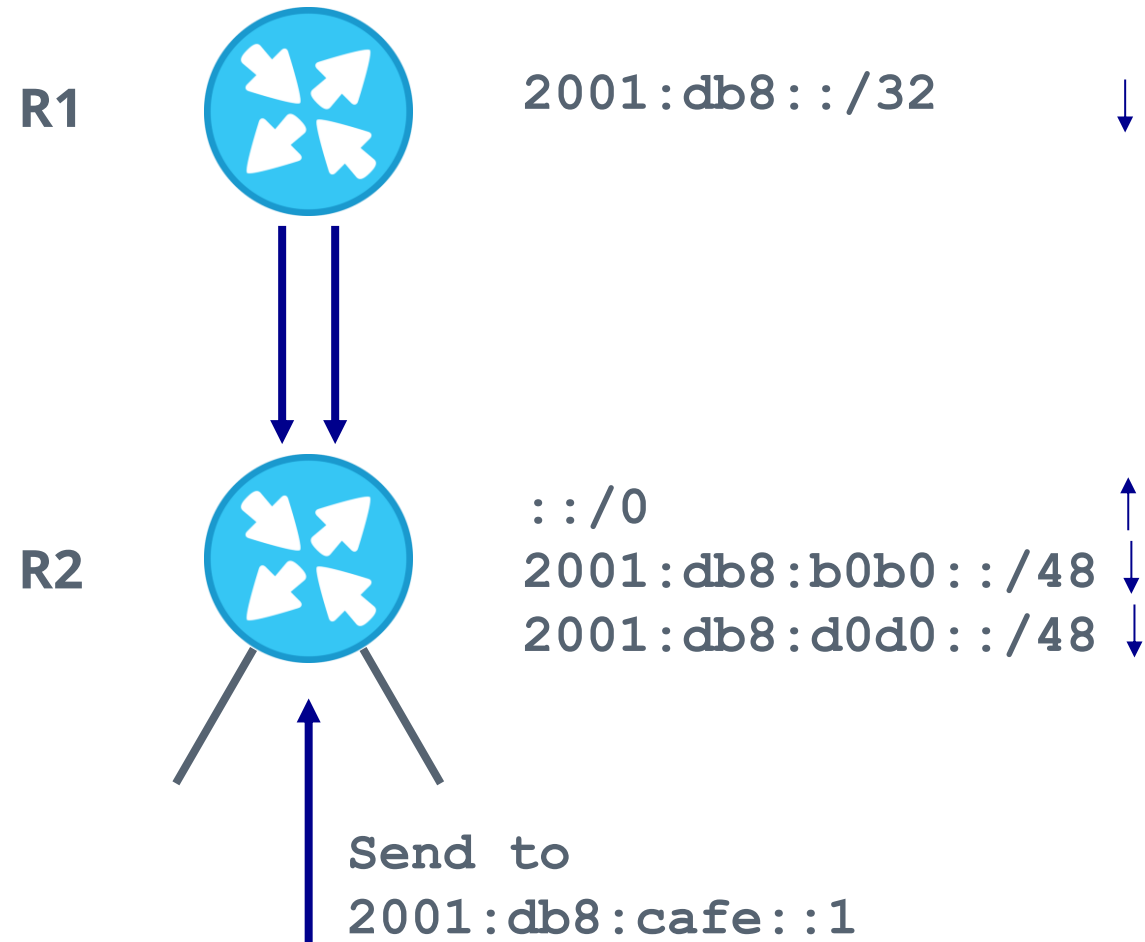
send to the source of the original trigger.



# A loop ultimately leads to an ICMP TTL Exceeded

In our case, not a single message but >250k. Why?

**A router bug leads to duplication of an ICMP Echo.**

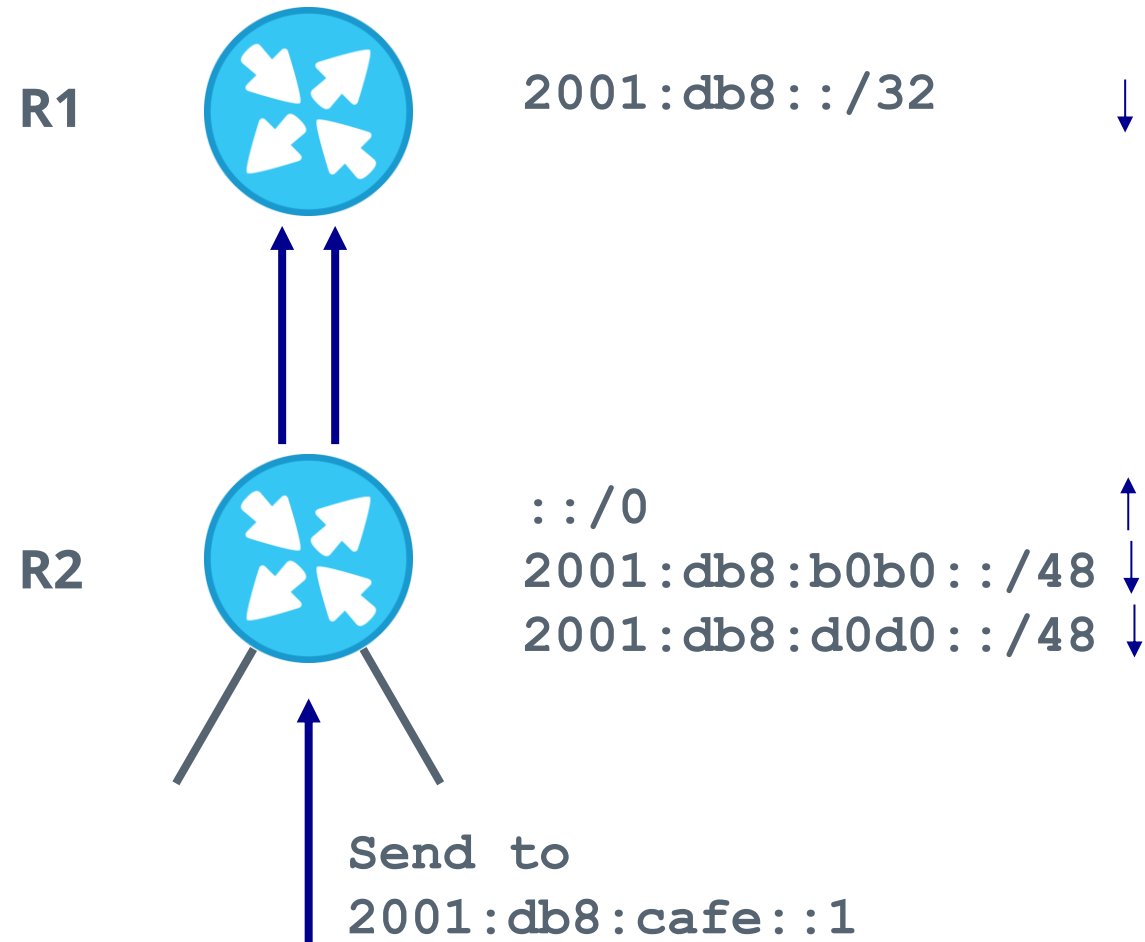


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**Each duplicated ICMP Echo is duplicated again.** Exponential increase of ICMP Echos between routers.

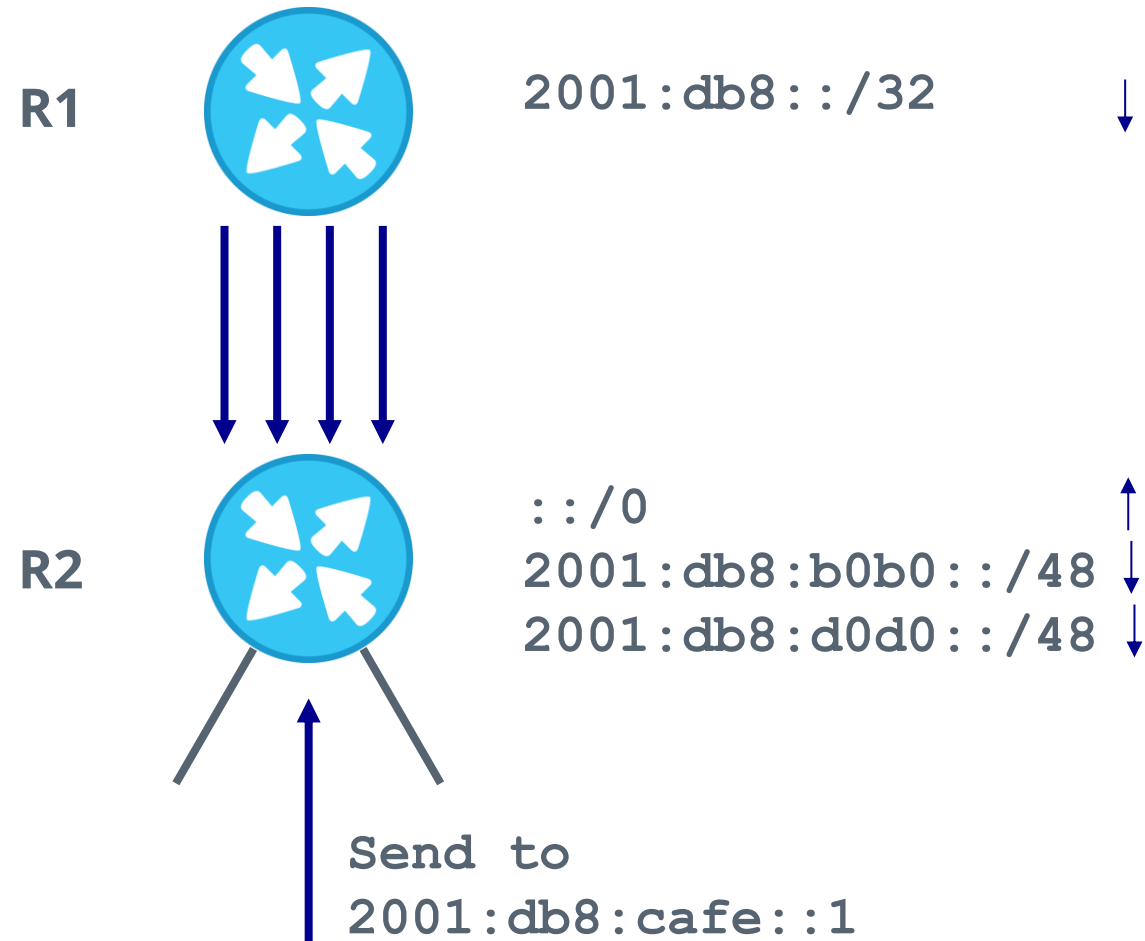


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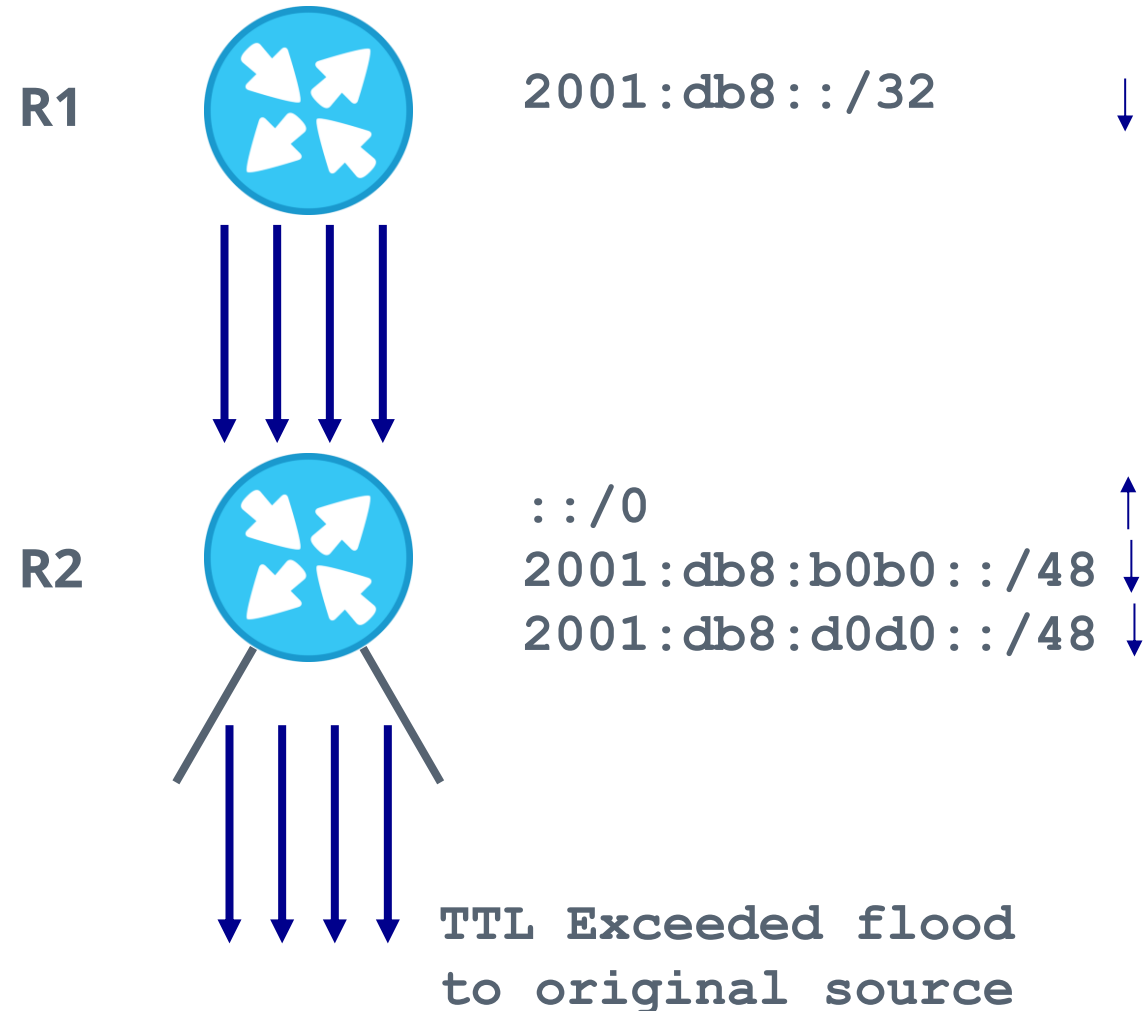
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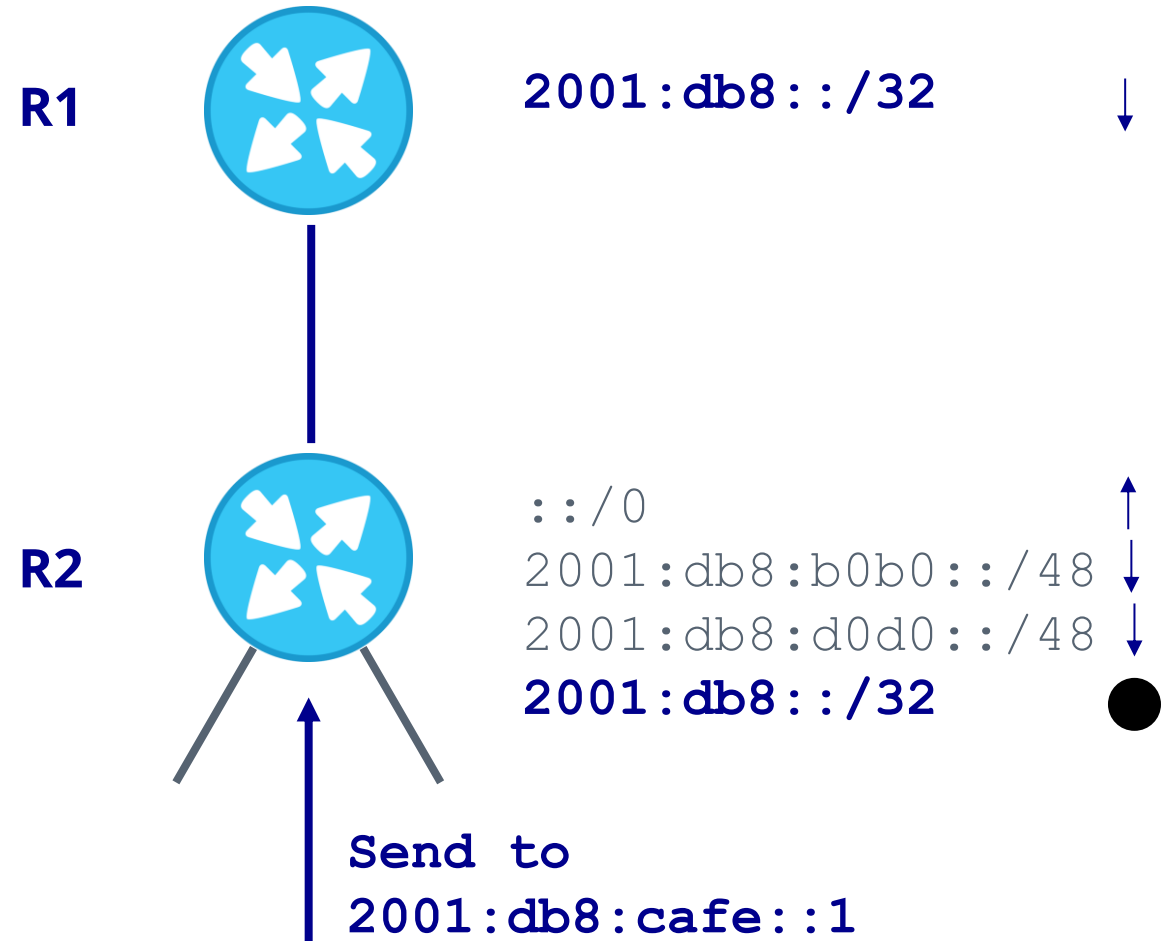
**Each duplicate triggers an individual TTL Exceeded.**

**Amplification at its best!**

[Confirmed by Juniper.]



# Preventing routing loops: Null route.



# How many IPv6 routing loops occur?

**November 2024**

**141M  
/48 subnets**

**March 2025**

**162M (+15%)  
/48 subnets**

# More details

## Lecture Internet Measurements

### Scanning the IPv6 Internet Using Subnet-Router Anycast Probing

MAYNARD KOCH, TU Dresden, Germany  
RAPHAEL HIESGEN, HAW Hamburg, Germany  
MARCIN NAWROCKI, NETSCOUT, USA  
THOMAS C. SCHMIDT, HAW Hamburg, Germany  
MATTHIAS WÄHLISCH, TU Dresden, Germany

Identifying active IPv6 addresses is challenging. Various methods emerged to master the measurement challenge in this huge address space, including hitlists, new probing techniques, and AI-generated target lists. In this short paper, we apply active Subnet-Router anycast (SRA) probing, a commonly unused method to explore the IPv6 address space. We compare our results with lists of active IPv6 nodes obtained from prior methods and with random probing. Our findings indicate that probing an SRA address reveals on average 10% more router IP addresses than random probing and is far less affected by ICMP rate limiting. Compared to targeting router addresses directly, SRA probing discovers 80% more addresses. We conclude that SRA probing is an important addition to the IPv6 measurement toolbox and may improve the stability of results significantly. We also find evidence that some active scans can cause harmful conditions in current IPv6 deployments, which we started to fix in collaboration with network operators.

CCS Concepts: • **Networks** → **Network measurement**; **Network structure**; **Network security**.

Additional Key Words and Phrases: IPv6, BGP, scanning, routing loops, subnet-router anycast

#### ACM Reference Format:

Maynard Koch, Raphael Hiesgen, Marcin Nawrocki, Thomas C. Schmidt, and Matthias Wählisch. 2025. Scanning the IPv6 Internet Using Subnet-Router Anycast Probing. *Proc. ACM Netw.* 4, CoNEXT3, Article 50 (December 2025), 12 pages. <https://doi.org/10.1145/3768997>

<https://doi.org/10.1145/3768997>

# The Open DNS

# When you operate DNS resolver infrastructure, you have two options

**Public**

**Any host on the Internet** should be able to send a request and get a reply.

**Close**

**Only hosts of your network** should be able to send a request and get a reply.

# When you operate DNS resolver infrastructure, you have two options



**Public**

**Any host on the Internet** should be able to send a request and get a reply.

**Examples of common public resolvers** are 8.8.8.8 and 9.9.9.9.

Any advantages of using public resolvers?

# Do you see any disadvantages of public resolvers?

# DNS Distributed Denial-of-Service (DDoS) Attacks are a Common Threat

radware SOLUTIONS PRODUCTS PARTNERS COMPANY RESOURCES SUPPORT

BLOG / DDOS PROTECTION / DNS UNDER SIEGE: REAL-WORLD DNS FLOOD ATTACKS

## DNS Under Siege: Real-World DNS Flood Attacks

By Sharon Shitrit October 25, 2015

How do real world DNS Flood look like? What is the impact on your DNS infrastructure and what can you do to boost your DNS security posture?

2023 DNS Flood Trends

We noted a considerable application-layer attack requests—have increased 2013 reaching a rate of traffic volume of these aimed to overload the common DNS query IPv4 query, account OTHER, and AAAA

### Someone Just Tried to Take Down Internet's Backbone with 5 Queries/Sec

Dec 10, 2015 Swati Khandelwal

#### The Internet's Backbone

## DNS Root Servers Hit by a Massive Cyber Attack

Someone just DDoSed one of the most critical organs of the Internet anatomy – The Internet's DNS Root Servers.

Early last week, a flood of as many as 5 Million queries per second hit many of the Internet's DNS (Domain Name System) Root Servers that act as the authoritative reference for mapping domain names to IP addresses and are a total of 13 in numbers.

The attack, commonly known as Distributed Denial of Service (DDoS) attack, took place on two separate occasions.

The first DDoS attack to the Internet's backbone root servers launched on November 30 that lasted 160

CATEGORIES: APT, AP, CU, D, P, T

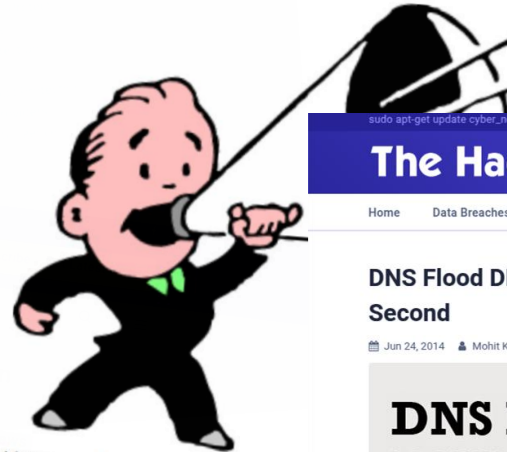
Developers Product News Security Radar Policy & Legal Zero Trust Speed & Reliability Life at Cloudflare Partners

## Deep Inside a DNS Amplification DDoS Attack

2012-10-30

Matthew Prince

5 min read



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### New DNS Vulnerability Lets Attackers Launch Large-Scale DDoS Attacks

May 20, 2020 Ravie Lakshmanan

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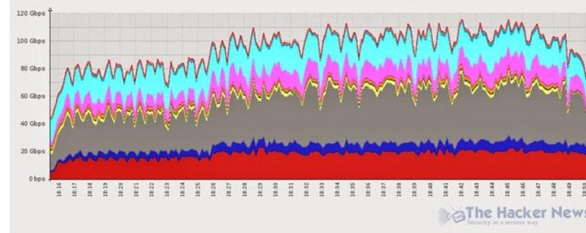
- Your Analysts Deserve More Than Alert Fatigue—Give Them an Agentive AI SOC Analyst
- Give CTEM a Look - You'll Never Want to Go Back to the Falling 10/60/90 Model
- Chainguard OS Reduces Supply

## DNS Flood DDoS Attack Hit Video Gaming Industry with 90 Million Requests per Second

Jun 24, 2014 Mohit Kumar

### DNS Flood DDoS Attack

90 Million requests/sec (Above 110 Gbps)

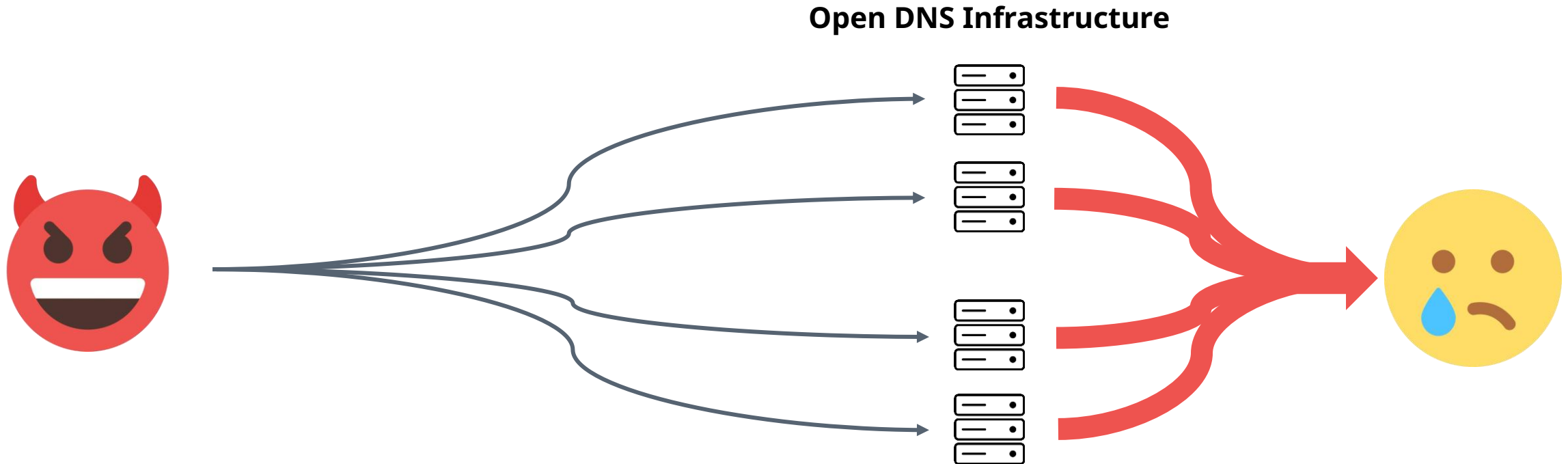


Hackers are leveraging large number of compromised machines (a botnet network) to carry out massive DNS Flood DDoS attack against a large Video Gaming Industry website, peaking above 110 Gbps.

A US based security solutions provider **Incapsula**, is protecting a famous Video Gaming website from this high bandwidth DDoS attack from last 48 Hours and according to them, the attack is still continuing. Incapsula has not yet disclosed the name of the video gaming company.

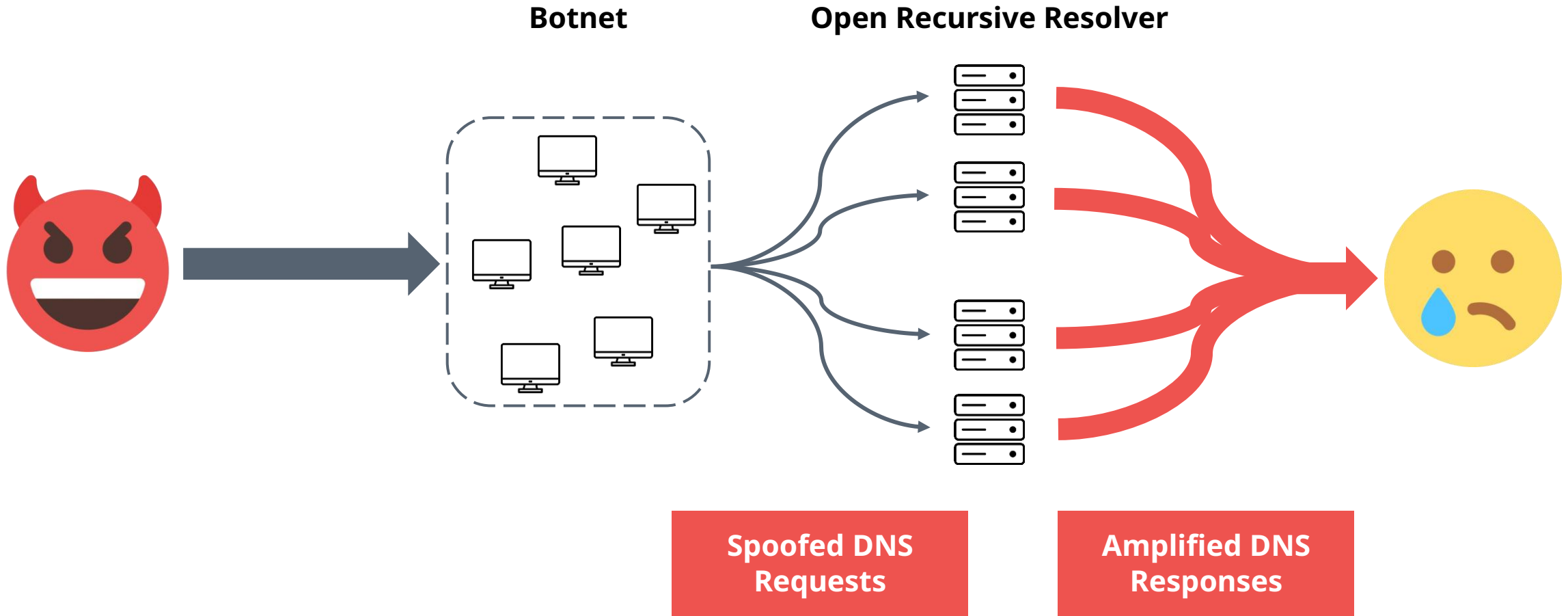
# The Open DNS (ODNS) Infrastructure is Misused in DDoS Attacks

Open components serve as reflectors or amplifiers.



# Potential Misuse Scenario of Open Recursive Resolvers

To run a distributed attack, the attacker needs access to a botnet.



# Potential Misuse Scenario of Open Recursive Resolvers

To run a distributed attack, the attacker needs access to a botnet.

Botnet

Open Recursive Resolver

**Botnets are costly.**

**Any alternatives?**

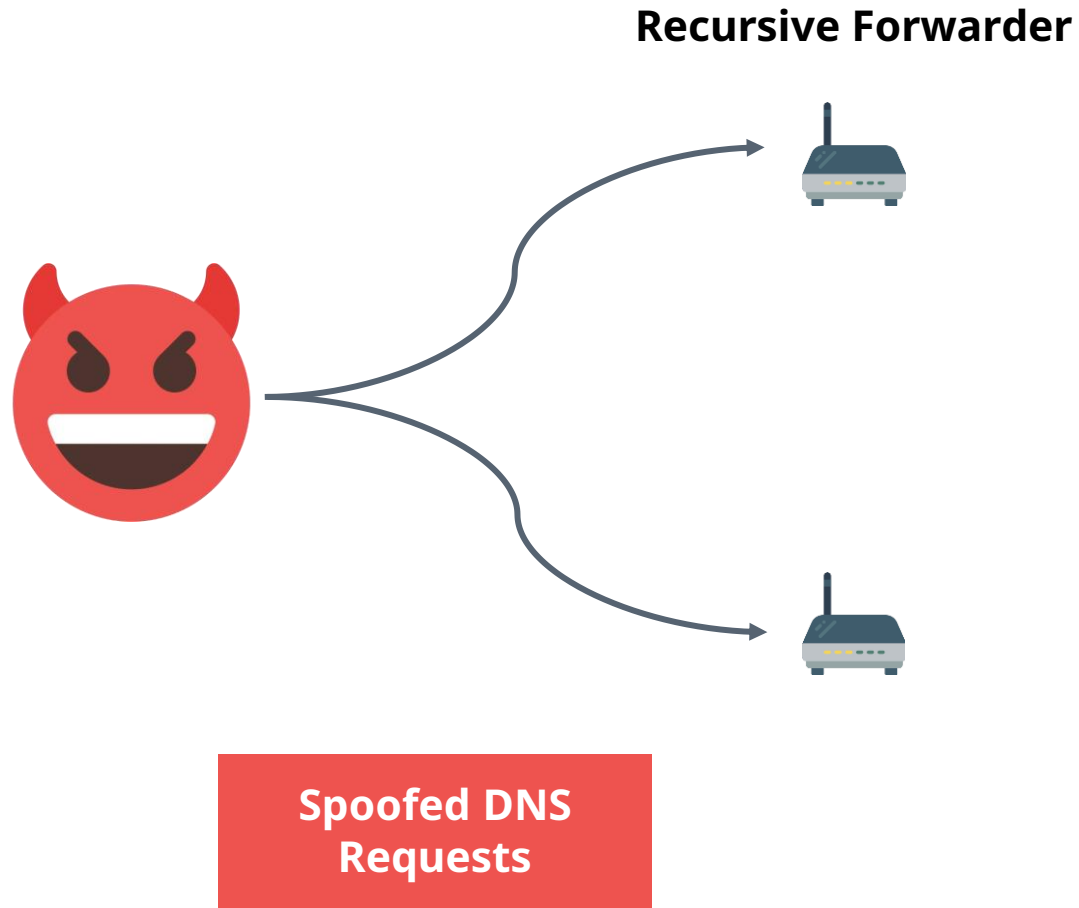


**Spoofed DNS  
Requests**

**Amplified DNS  
Responses**

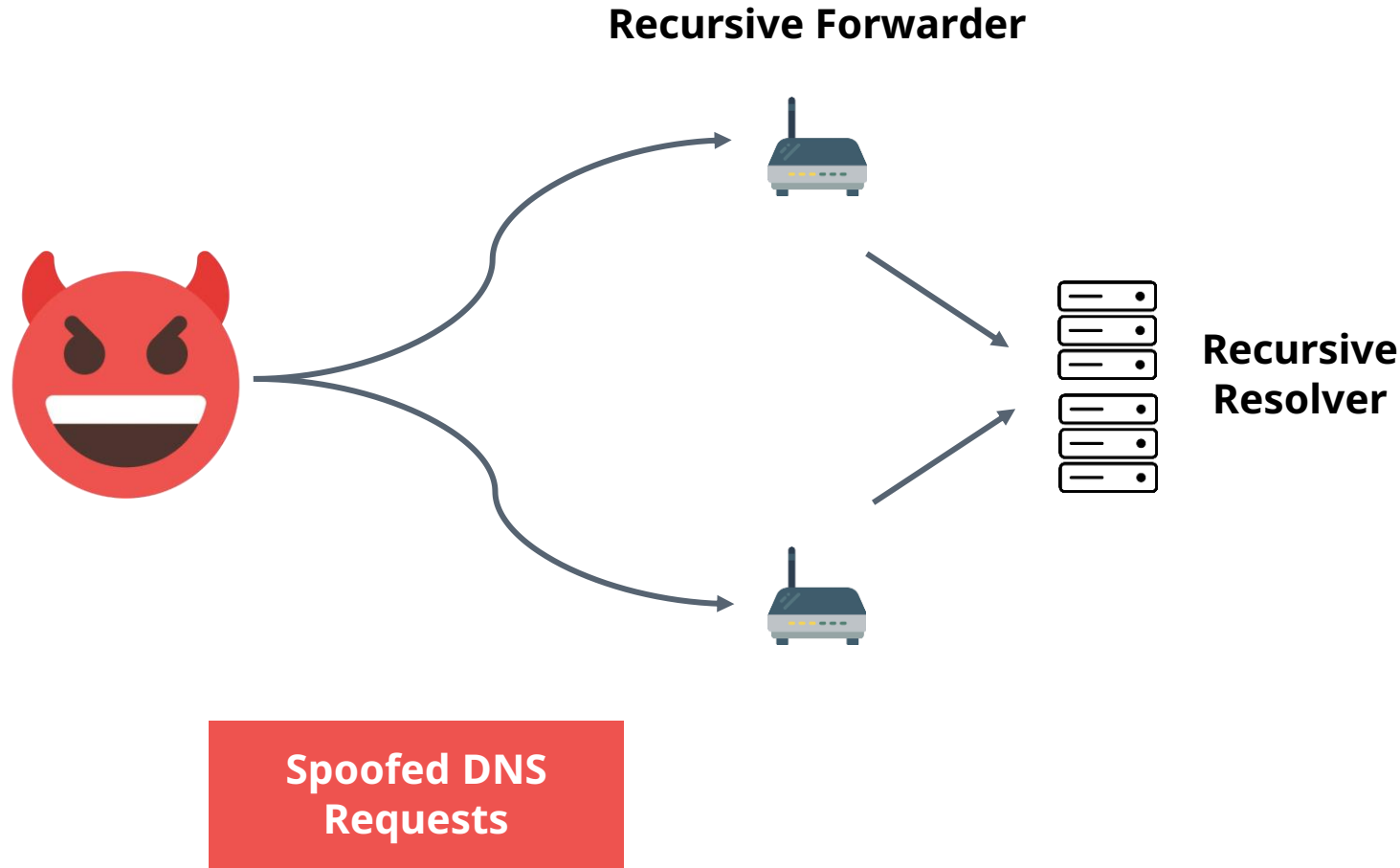
# Potential Misuse Scenario of Open Recursive Forwarders

Forwarders are globally distributed, enabling distributed attacks w/o a botnet.



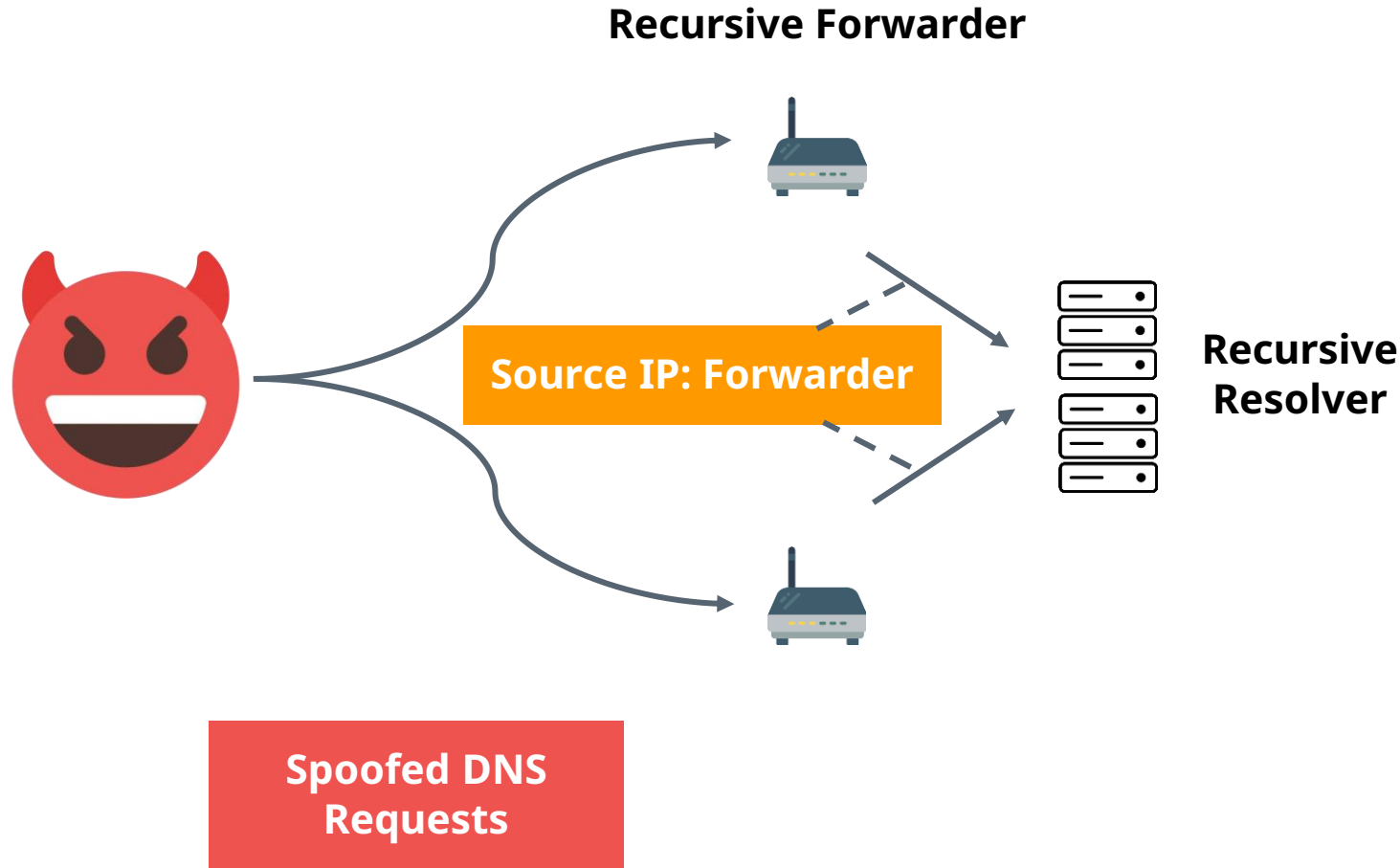
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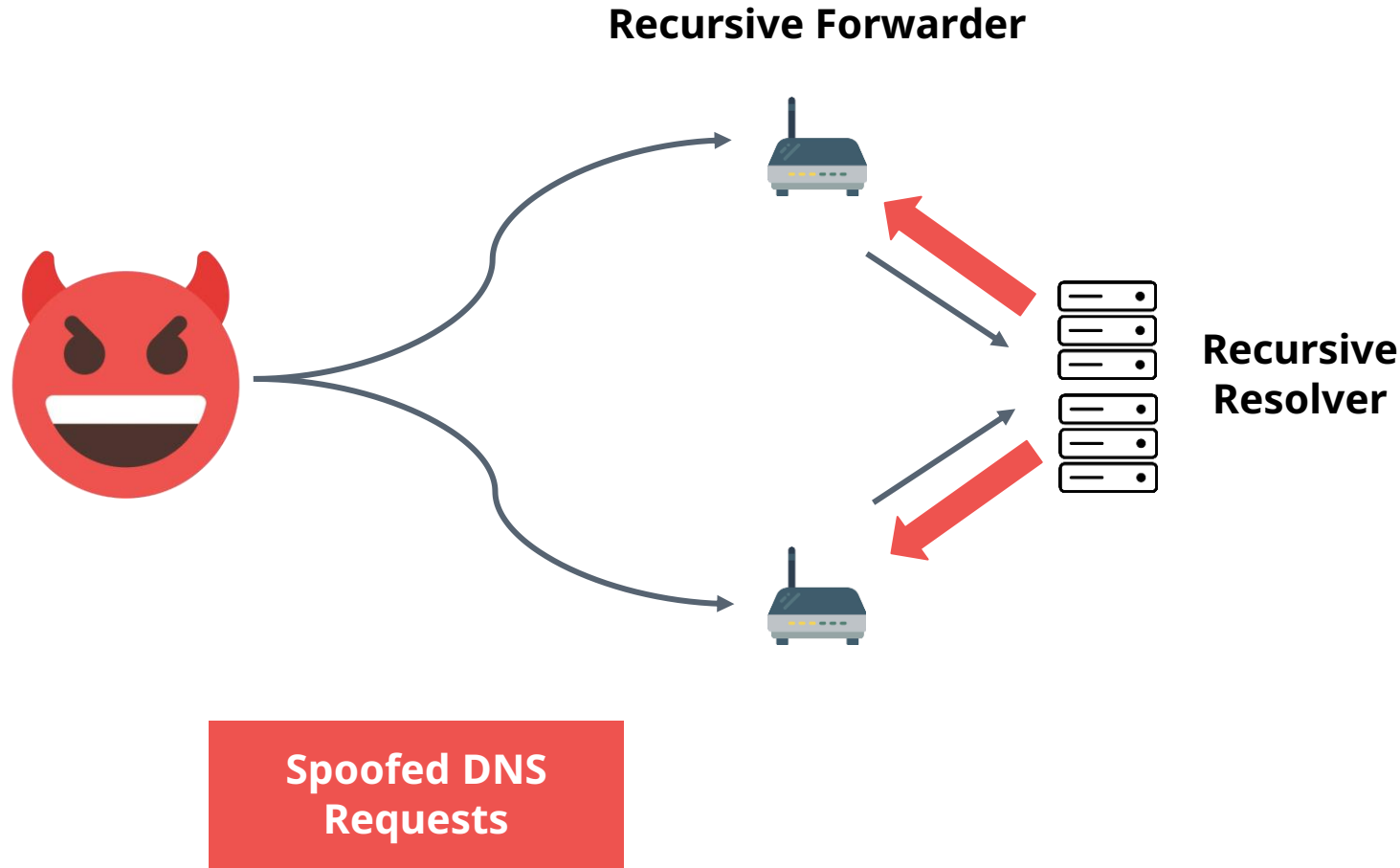
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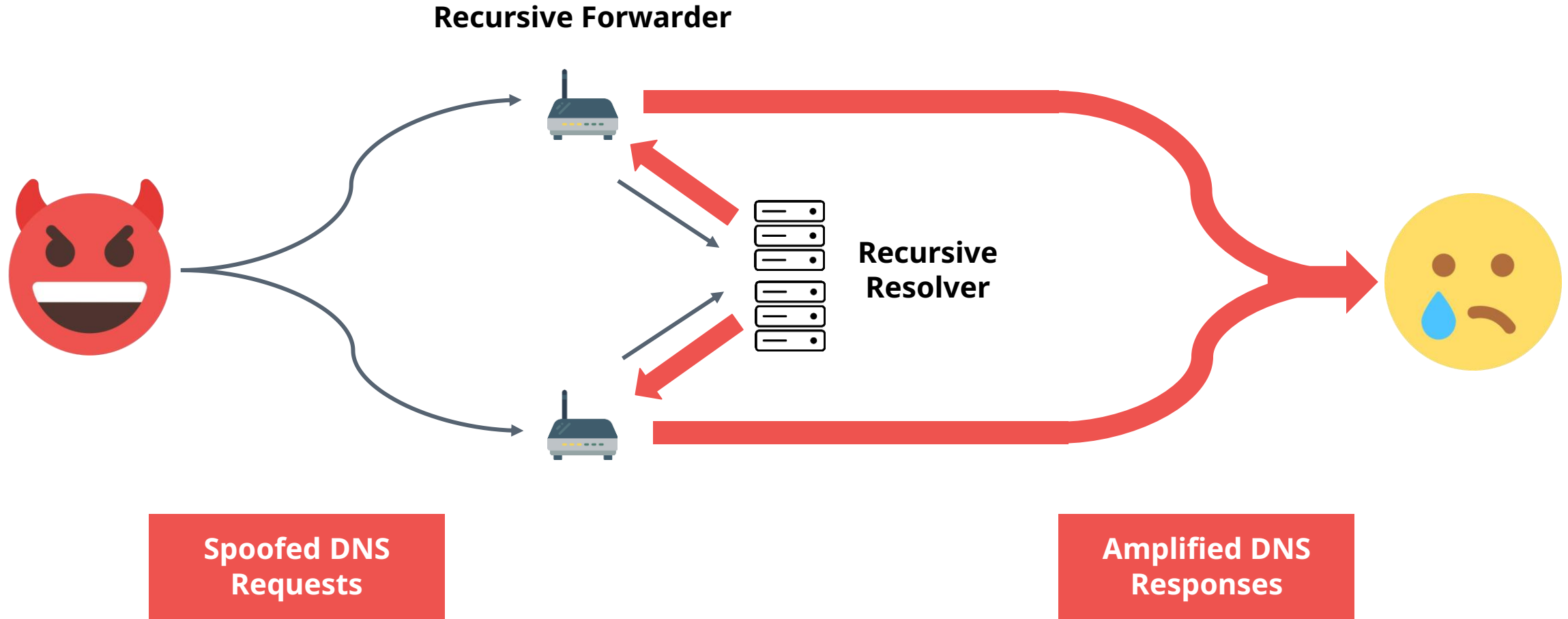
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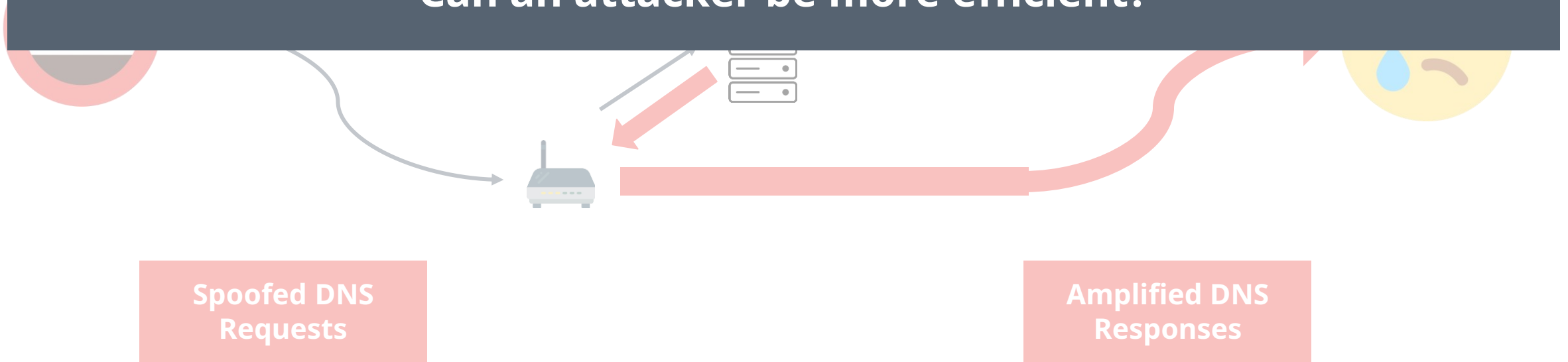
# Potential Misuse Scenario of Open Recursive Forwarders

Forwarders are globally distributed, enabling distributed attacks w/o a botnet.

Recursive Forwarder

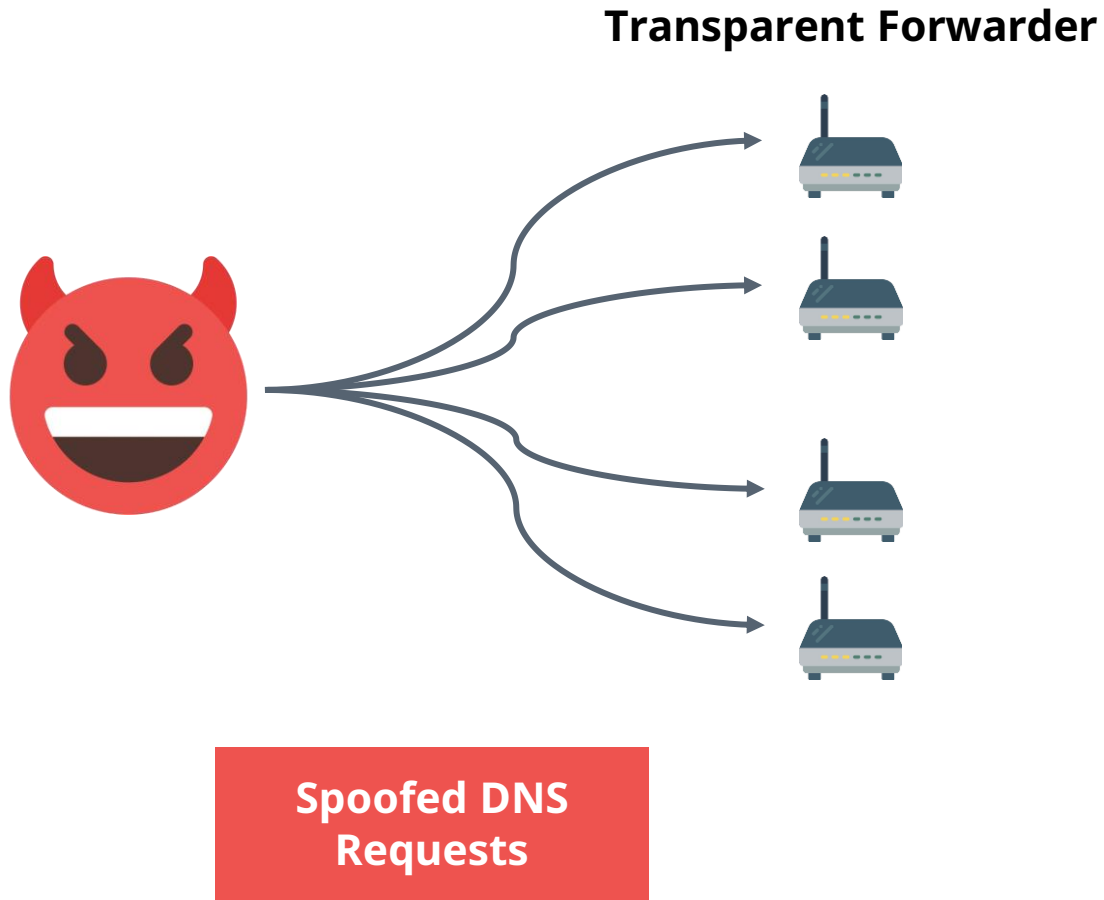
**A recursive forwarder has to handle the amplified response.**

**Can an attacker be more efficient?**



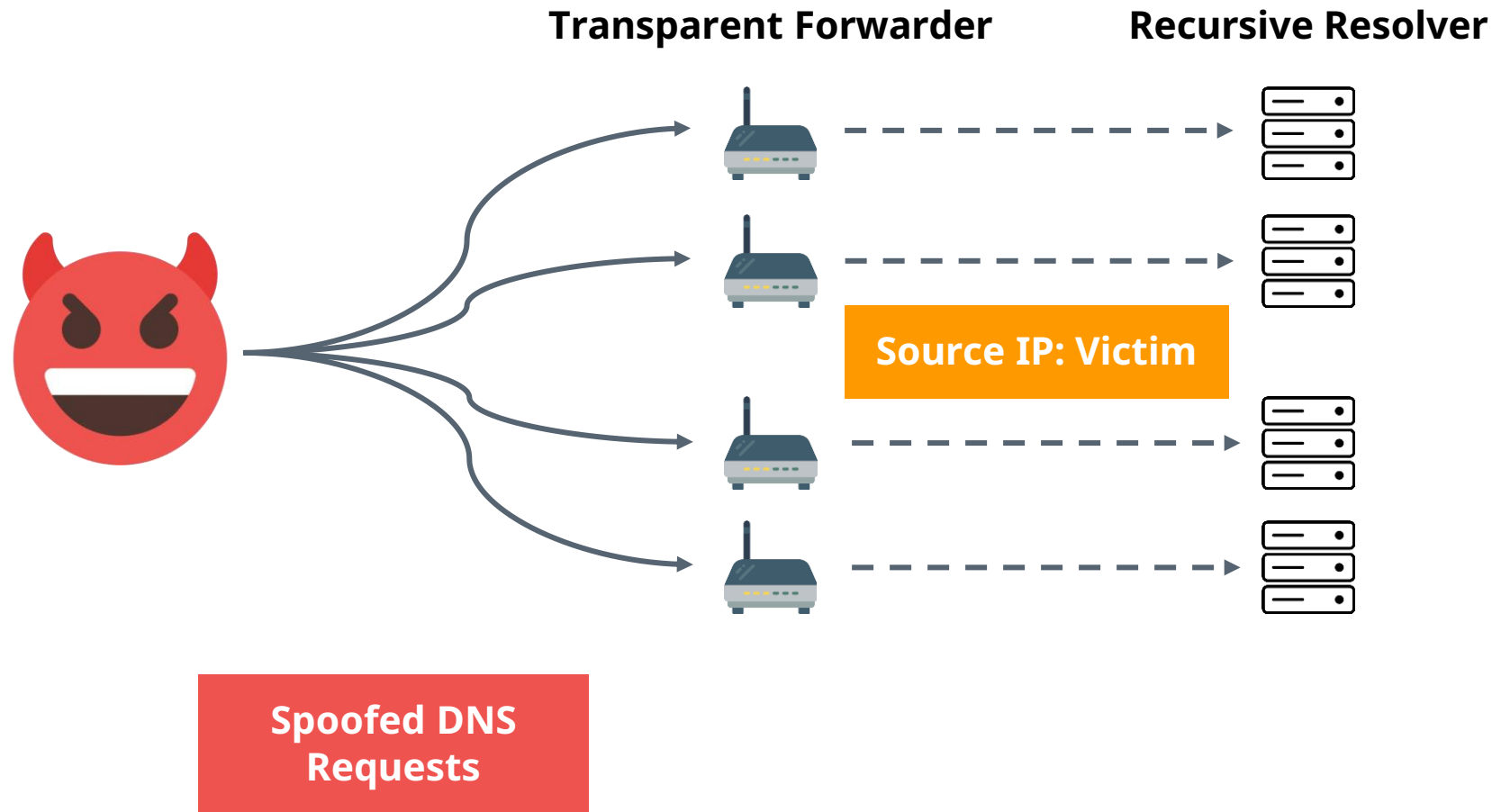
# Transparent DNS Forwarders

Misconfigured devices, not intentionally deployed.



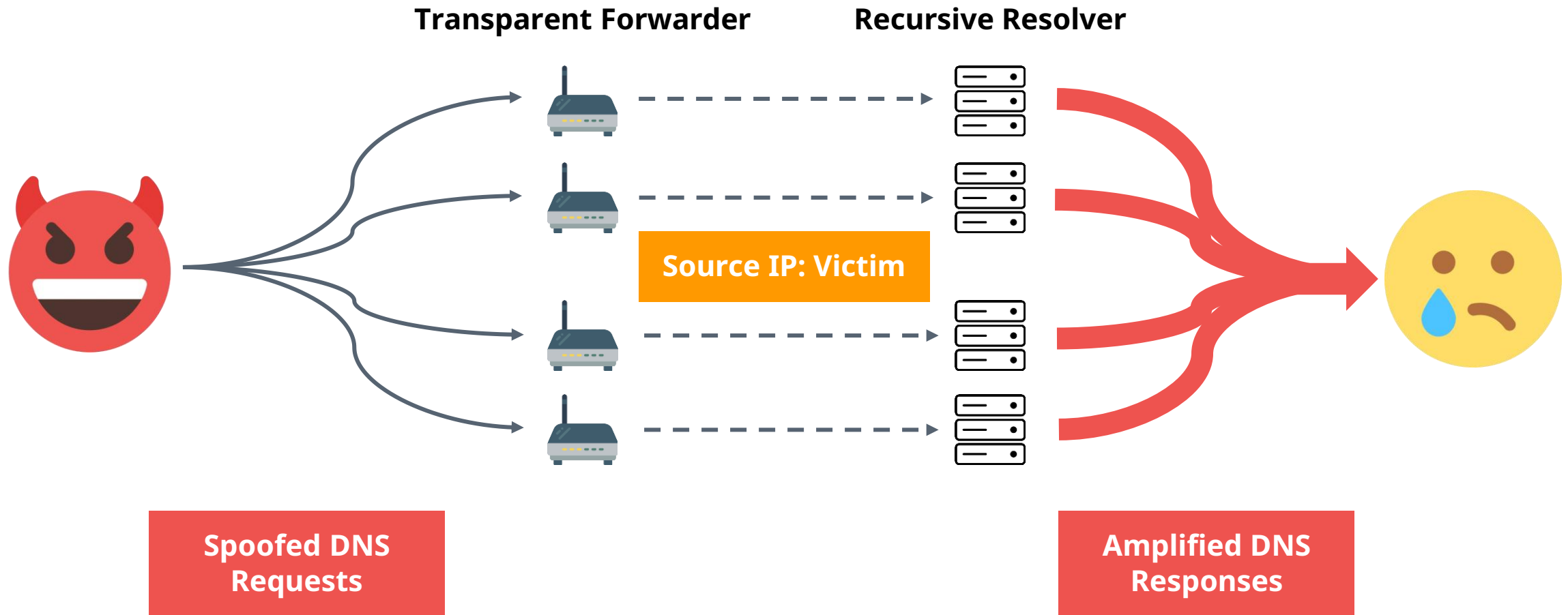
# Transparent DNS Forwarders

They do not rewrite the source IP address of the arriving packet.



# Transparent DNS Forwarders

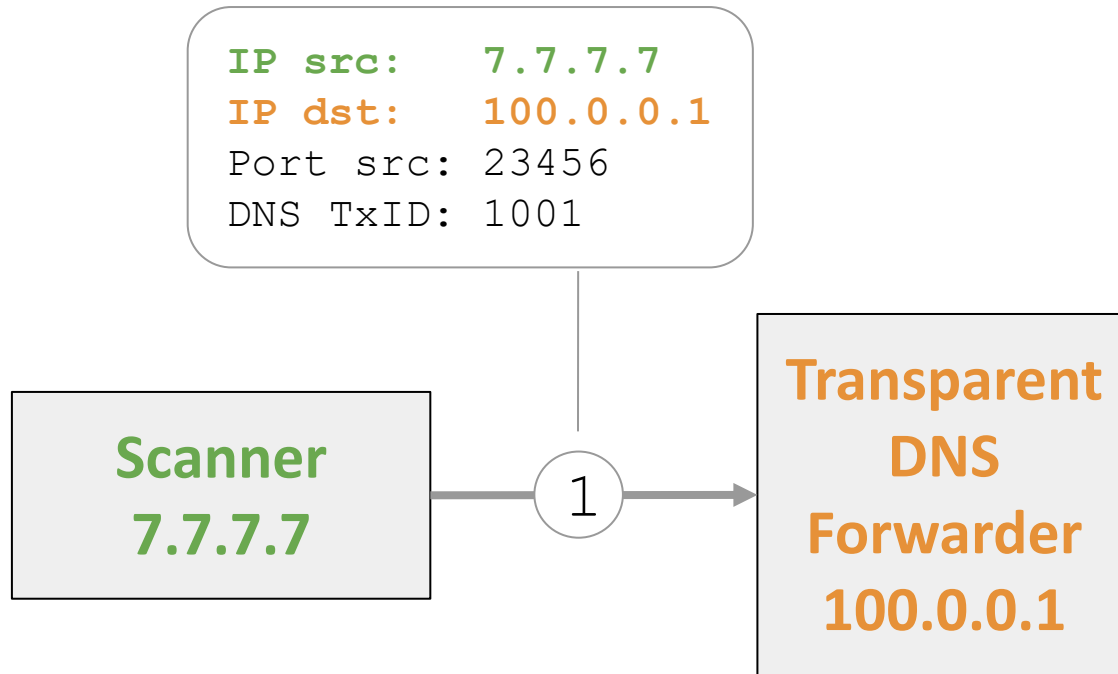
The recursive resolver allegedly receives a DNS request from the DDoS target.



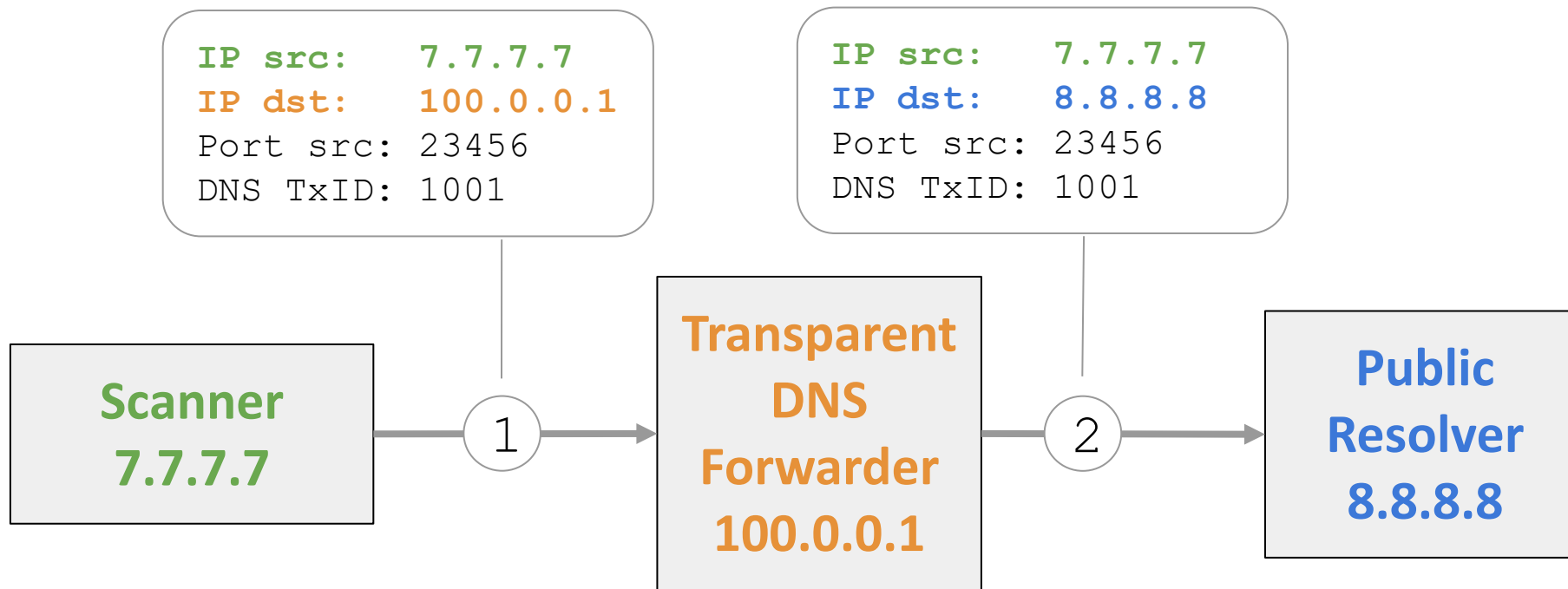
# Detailed example

**Transparent  
DNS  
Forwarder  
100.0.0.1**

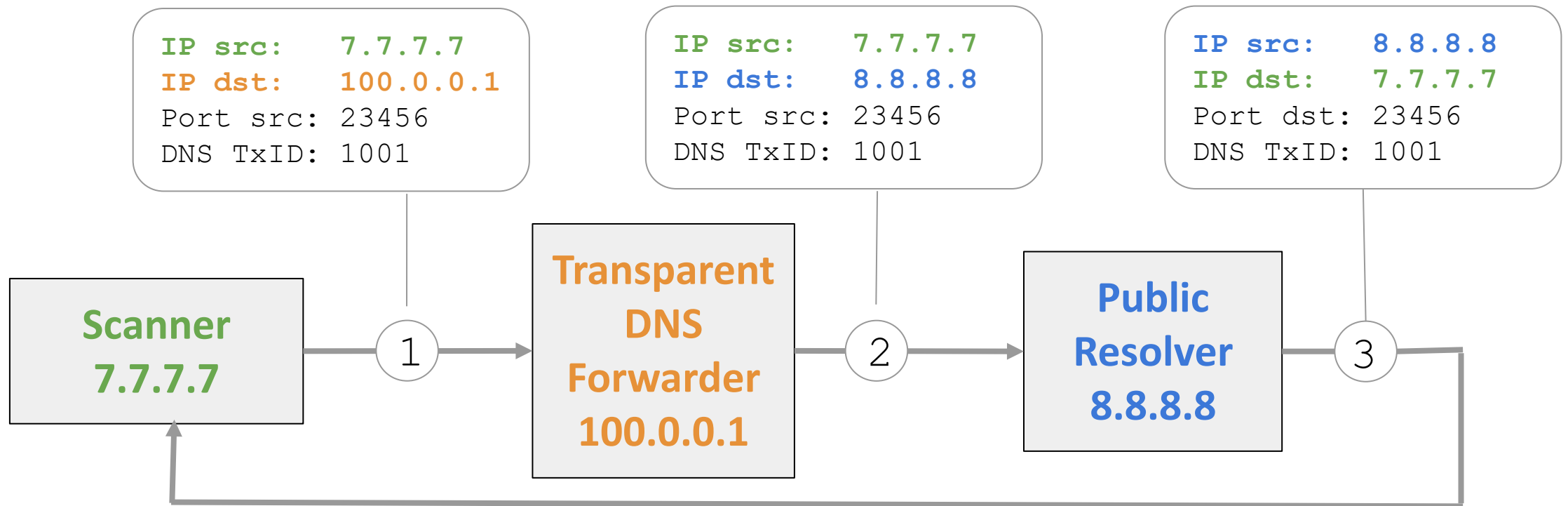
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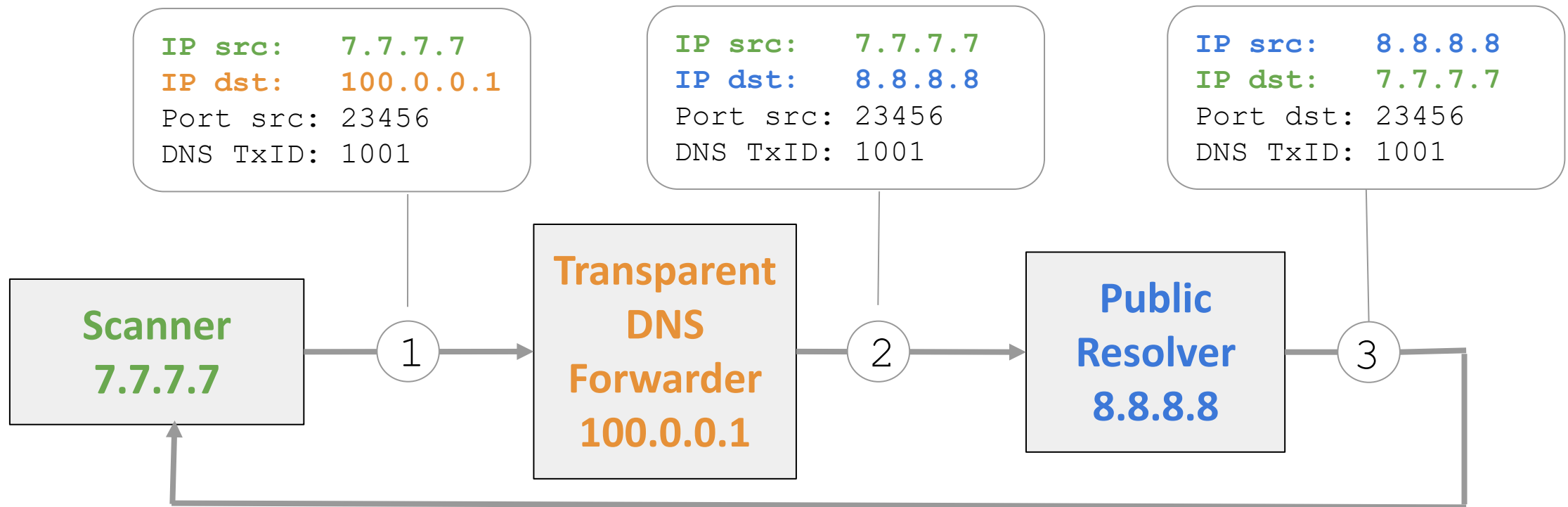
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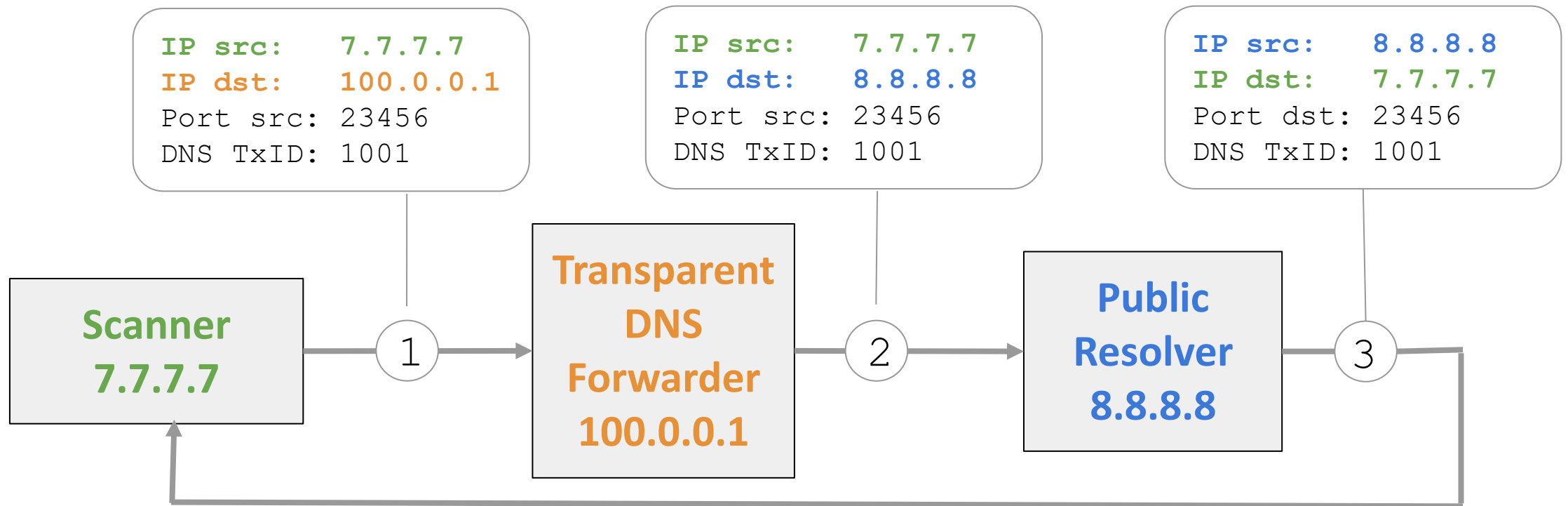


# Detailed example



Transparent forwarders  
send spoofed packets.

# Detailed example



Transparent forwarders  
send spoofed packets.

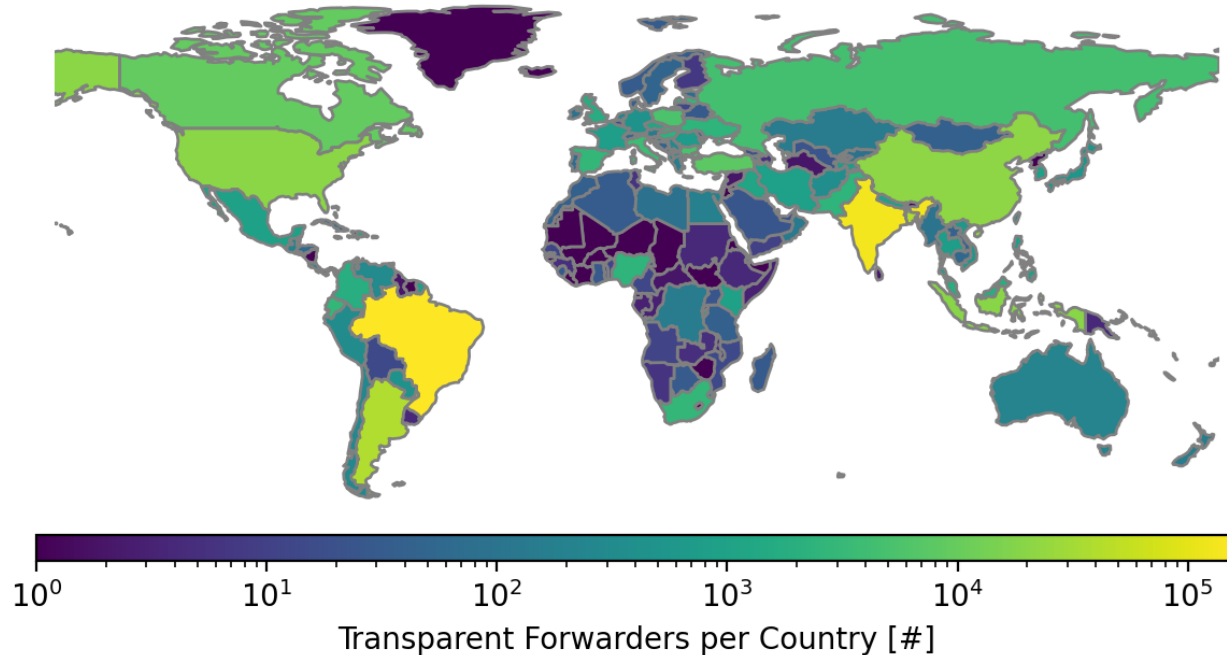
Scanners that consider only  
the replying source IP  
address make mistakes.

# We now focus on transparent forwarders

|                                 | Censys                                                                              | Shadowserver                                                                        | Shodan                                                                              | Our Scans                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| # ODNS                          | 1.75M                                                                               | 1.8M                                                                                | 1.6M                                                                                | 2.15M                                                                                                   |
| Transparent forwarders detected |  |  |  | <br>(26% forwarders) |

# Additional Attack Infrastructure – Geographical Distribution

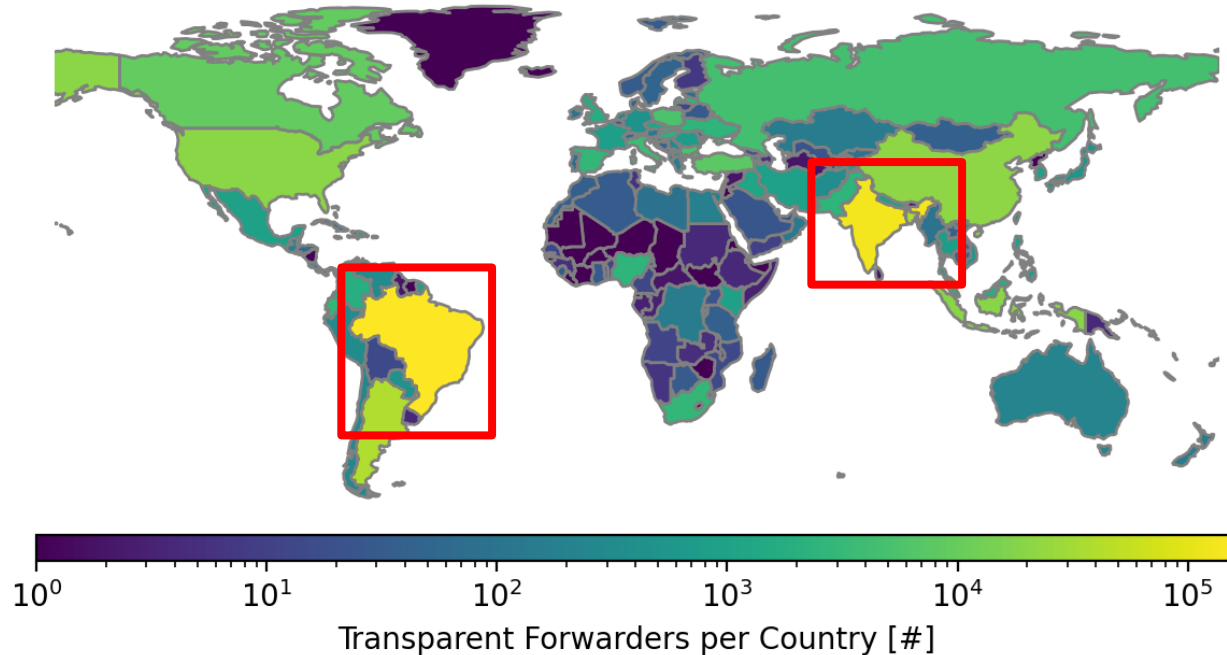
More than 30% of the open DNS infrastructure are transparent forwarders.



**Large number of devices. 540k of all open DNS components are transparent forwarders.**

# Additional Attack Infrastructure – Geographical Distribution

Brazil and India host >50% of all transparent forwarders.

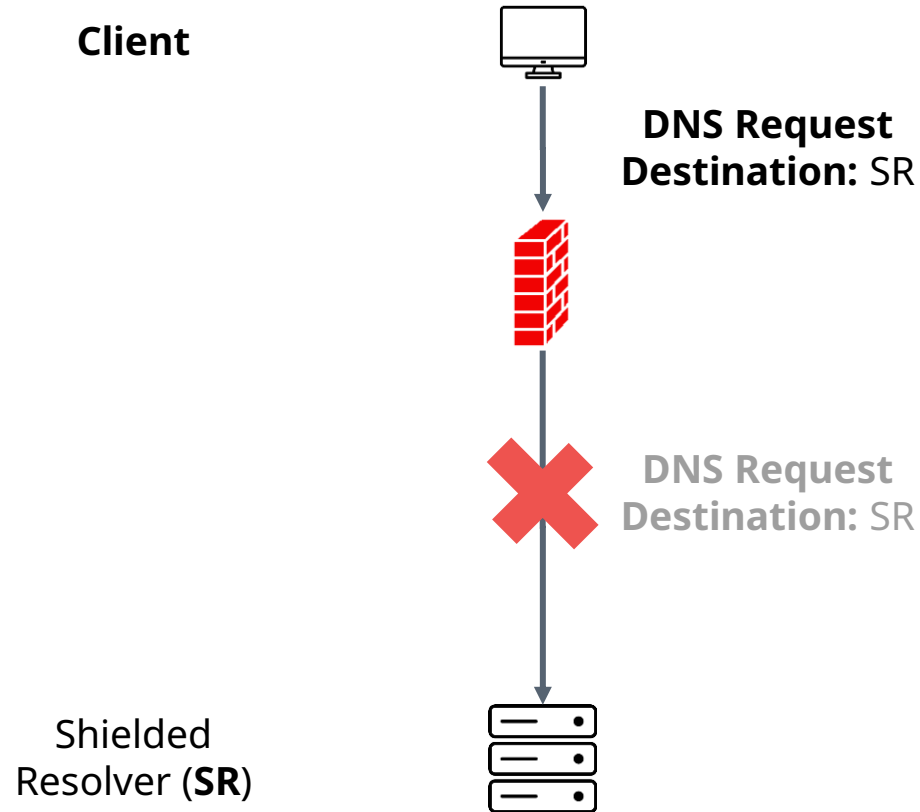


**Large number of devices. 540k** of all open DNS components are **transparent forwarders**.

**Widely distributed with regional bias.** Transparent forwarders can be found in **175 countries**, most of them are active in Brazil and India.

# Additional Attack Infrastructure – Shielded Resolvers

Transparent forwarders enable access to shielded resolvers.

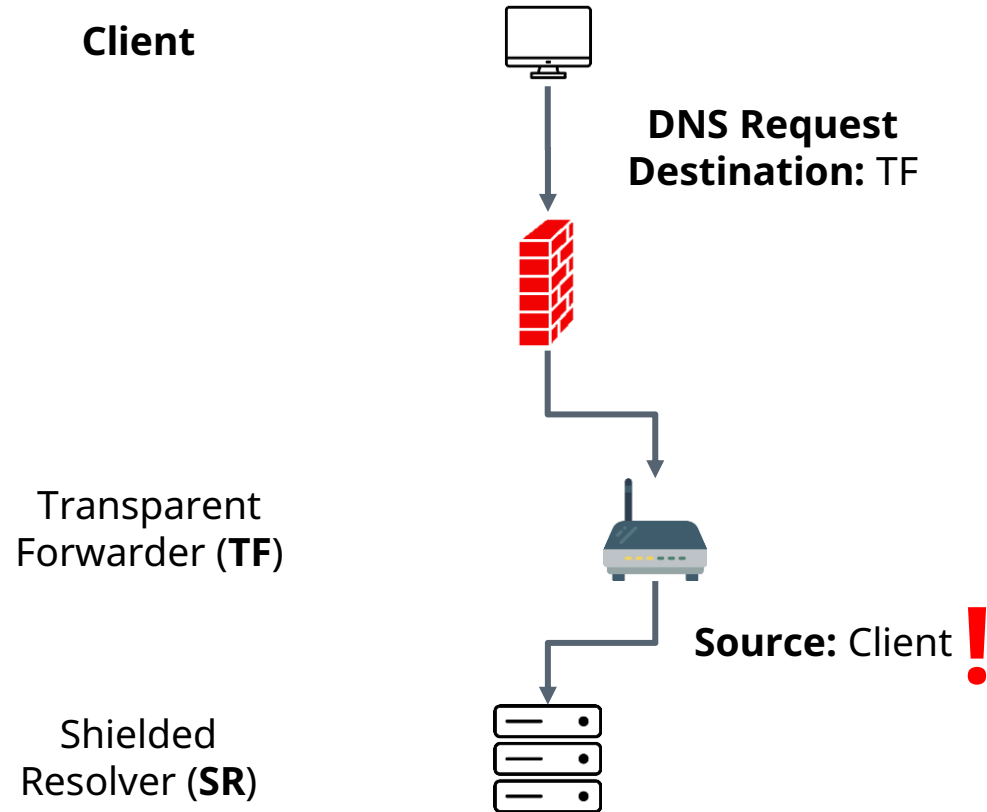


**Shielded resolvers** should only be accessible from the internal network.

**Firewall at the network border** filters requests that directly target the shielded resolver.

# Additional Attack Infrastructure – Shielded Resolvers

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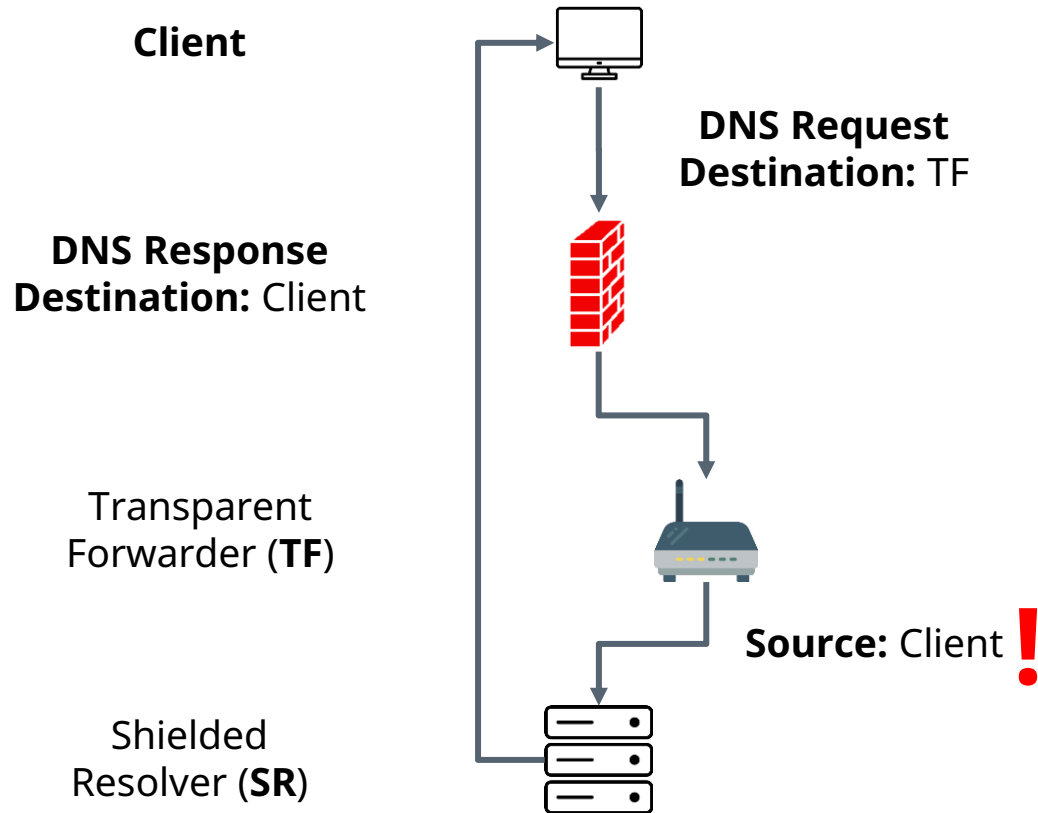
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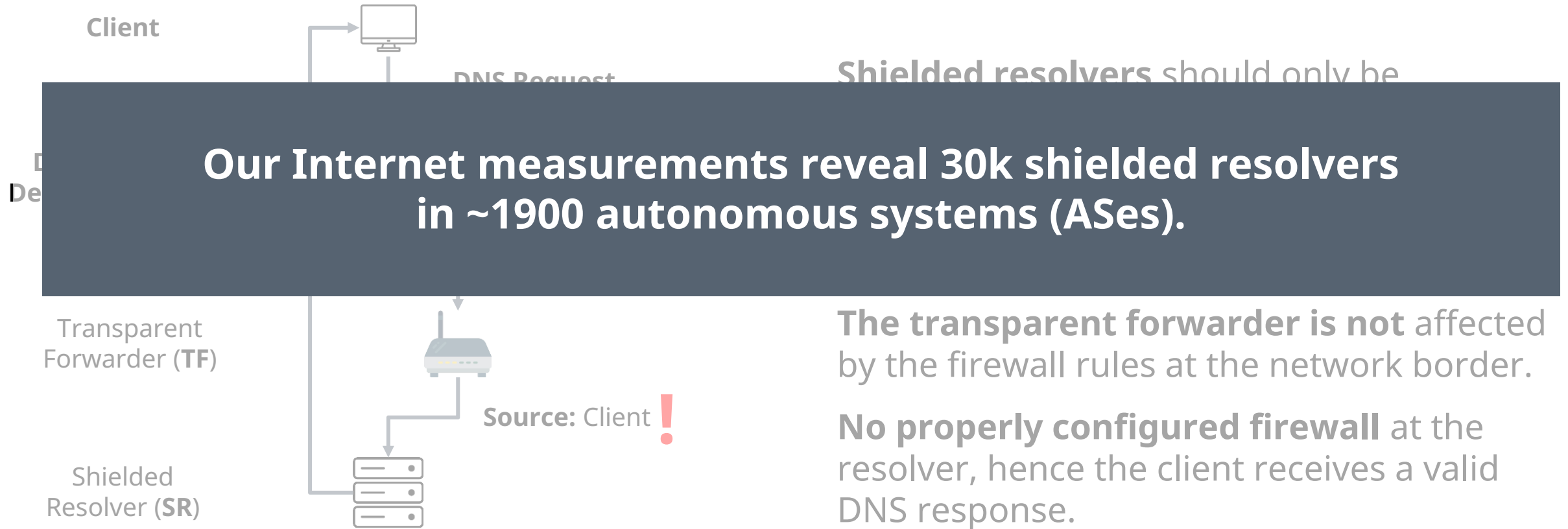
**Firewall at the network border** filters requests that directly target the shielded resolver.

**The transparent forwarder is not** affected by the firewall rules at the network border.

**No properly configured firewall** at the resolver, hence the client receives a valid DNS response.

# Additional Attack Infrastructure – Shielded Resolvers

Transparent forwarders enable access to shielded resolvers.



# Agenda

1. Additional Attack Infrastructure
2. **Higher Efficiency in Potential Attacks**
3. Avoiding Local Rate Limits by Orchestrating Transparent Forwarders
4. Responsible Disclosure and Mitigation Options

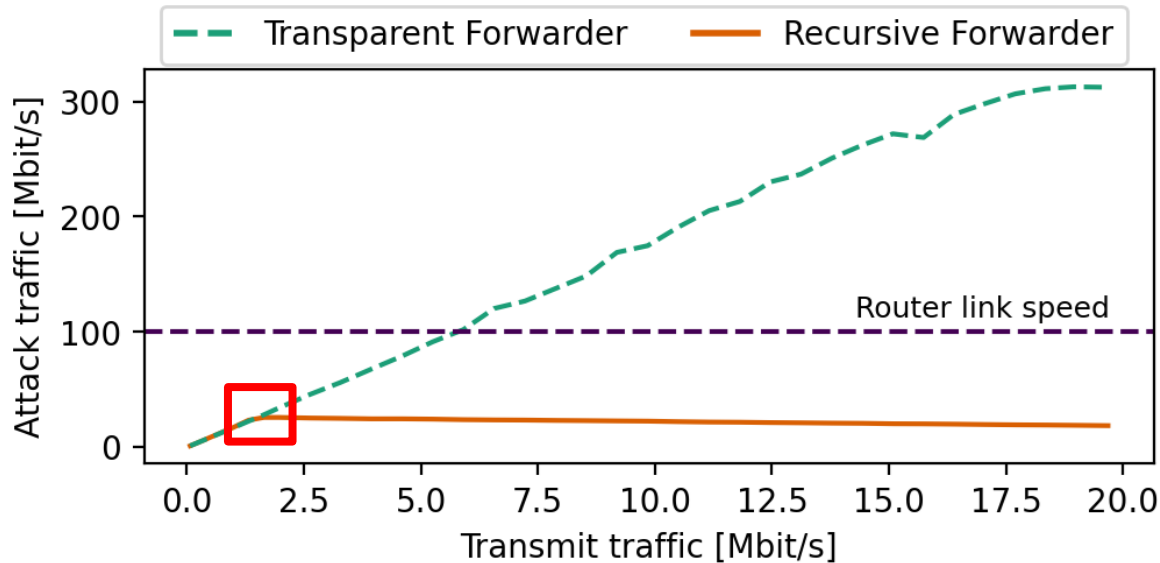
# Transparent Forwarders Allow For More Attack Traffic

They do not need to handle the amplified DNS response.

**Testbed.** We use a MikroTik RB750Gr3 router (i) as a transparent forwarder and (ii) as a recursive forwarder to show the advantage of transparent forwarders.

# Transparent Forwarders Allow For More Attack Traffic

Recursive forwarder quickly reaches its maximum attack rate.

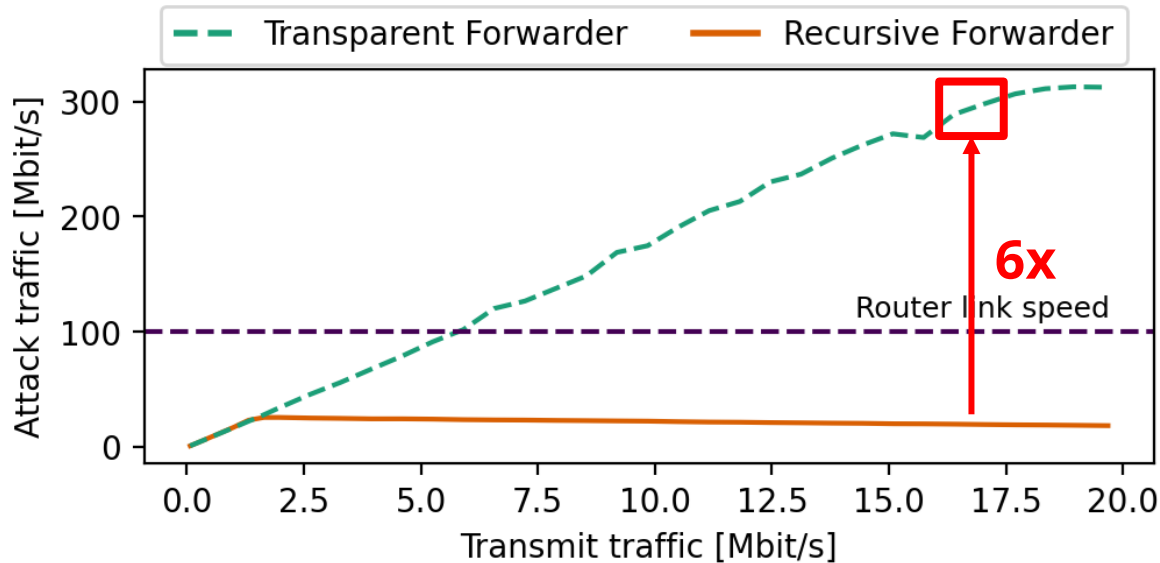


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**Recursive forwarder.** The router already runs into resource limitations at 1.5Mbit/s of query traffic, resulting in **≈50Mbit/s** of attack traffic.

# Transparent Forwarders Allow For More Attack Traffic

Transparent forwarder generates more than 300Mbit/s (6x) attack traffic.



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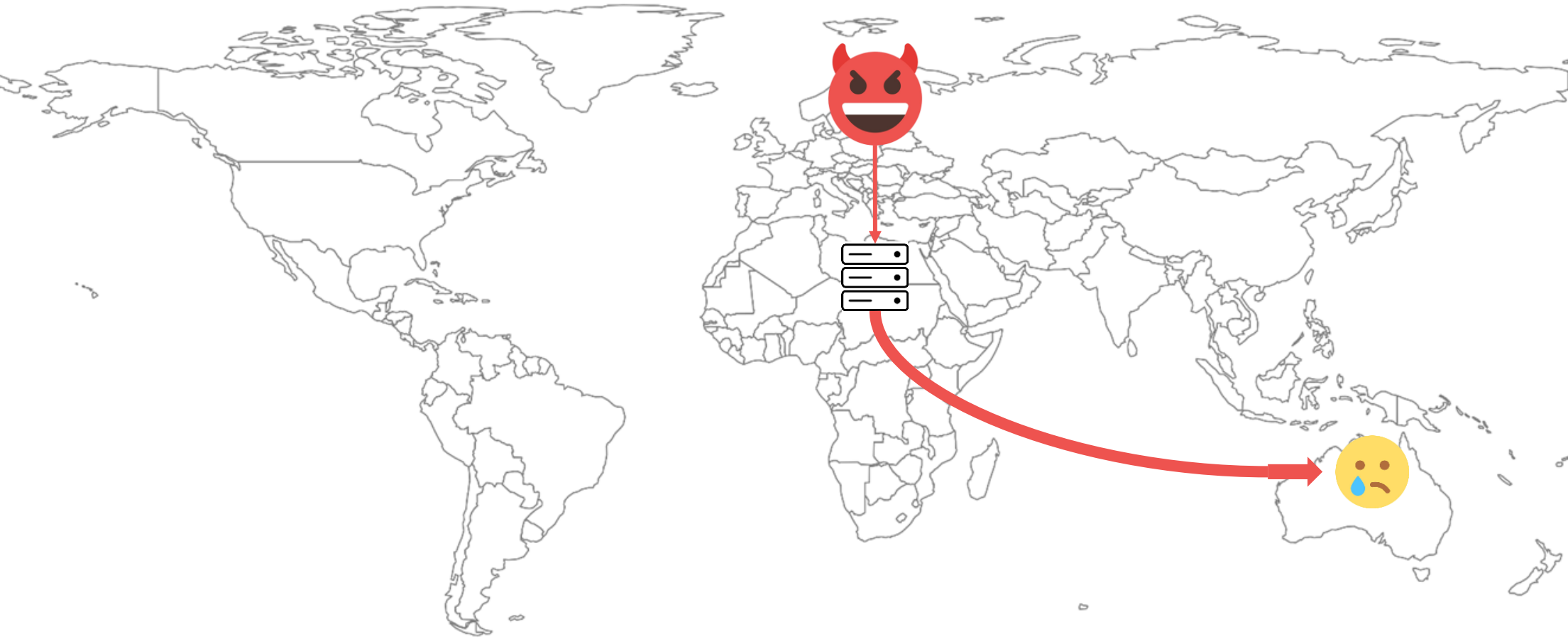
**Transparent forwarder.** We reach more than **300Mbit/s** of attack traffic before running into resource limitations of the MikroTik router.

# Agenda

1. Additional Attack Infrastructure
2. Higher Efficiency in Potential Attacks
3. **Avoiding Local Rate Limits by Orchestrating Transparent Forwarders**
4. Responsible Disclosure and Mitigation Options

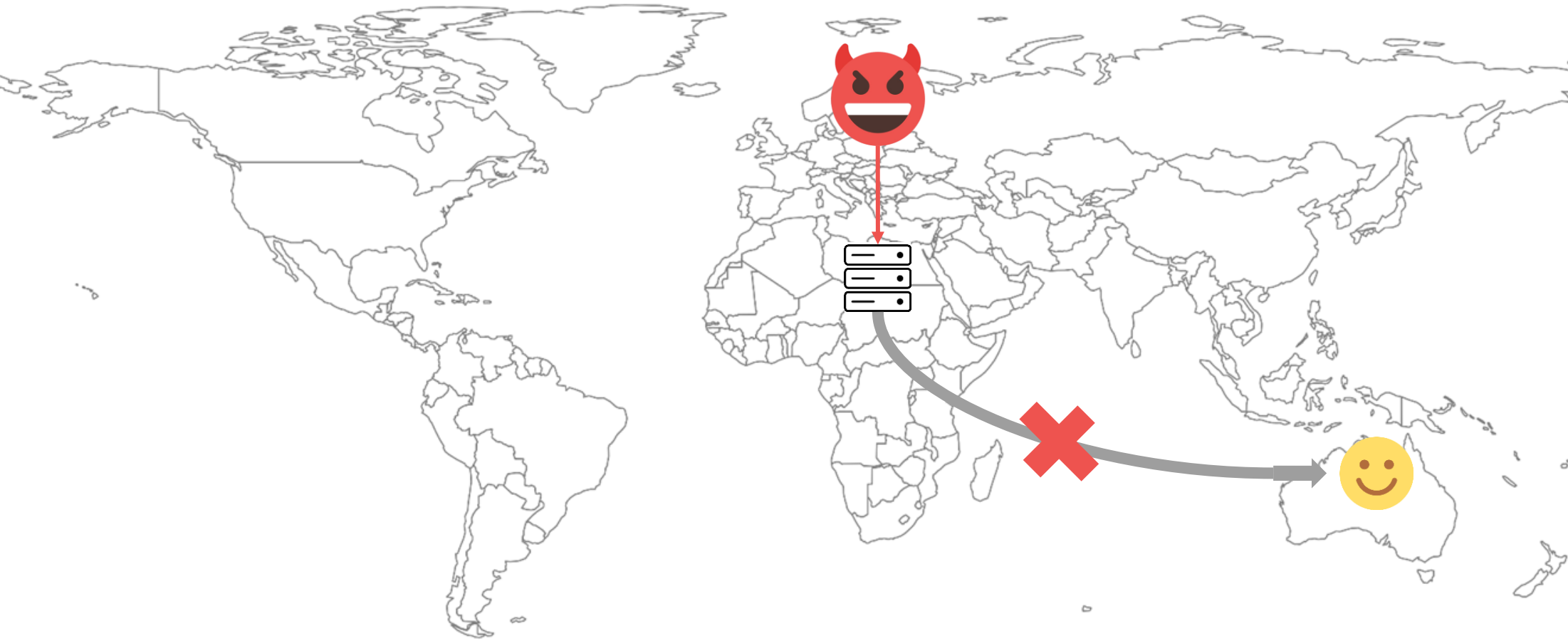
# The Attacker Controls a Single Vantage Point

Without a botnet, the attacker reaches only a single point of presence (PoP).



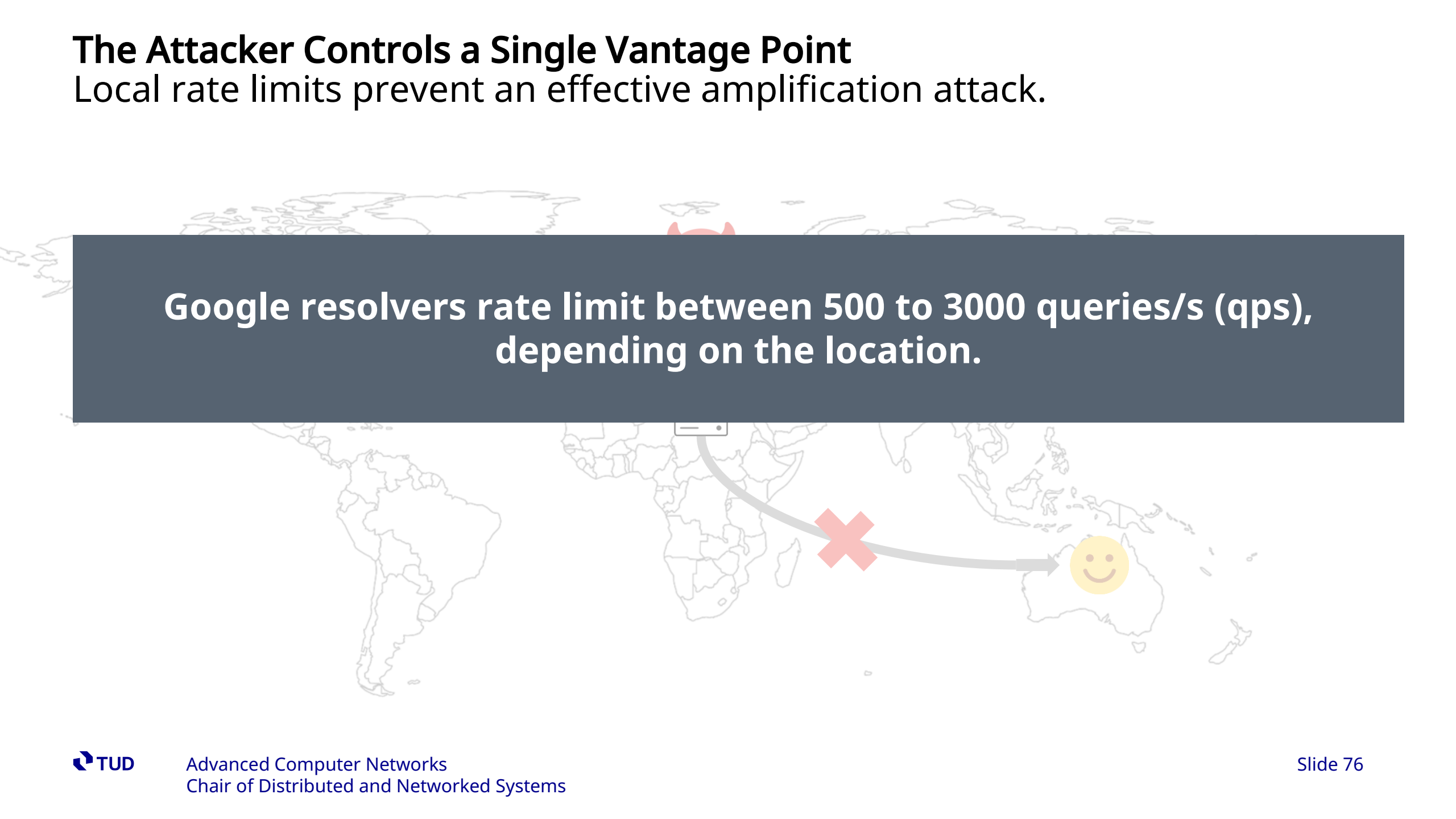
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Local rate limits prevent an effective amplification attack.



# The Attacker Controls a Single Vantage Point

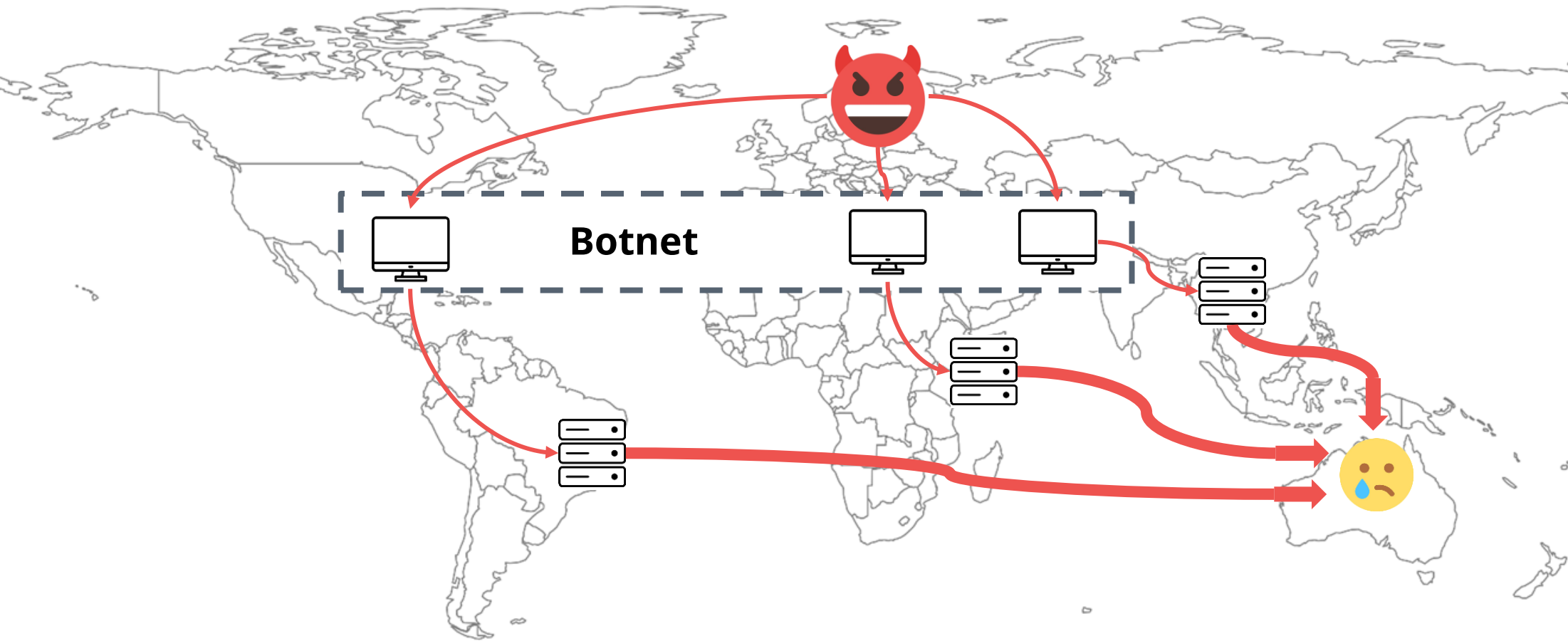
Local rate limits prevent an effective amplification attack.



**Google resolvers rate limit between 500 to 3000 queries/s (qps),  
depending on the location.**

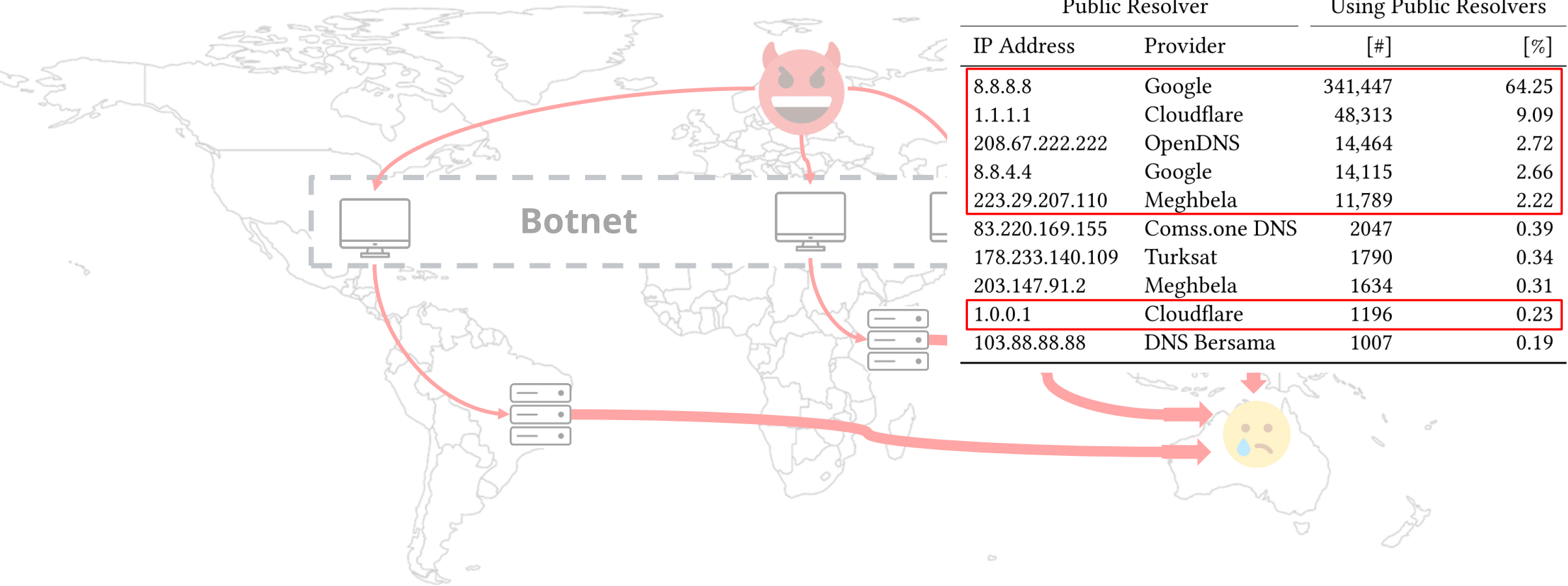
# The Attacker Controls a Botnet

Local rate limits can be avoided through global distribution.



# Transparent Forwarders Relay Requests to Anycast Deployments

An attacker can exploit the anycast infrastructure.



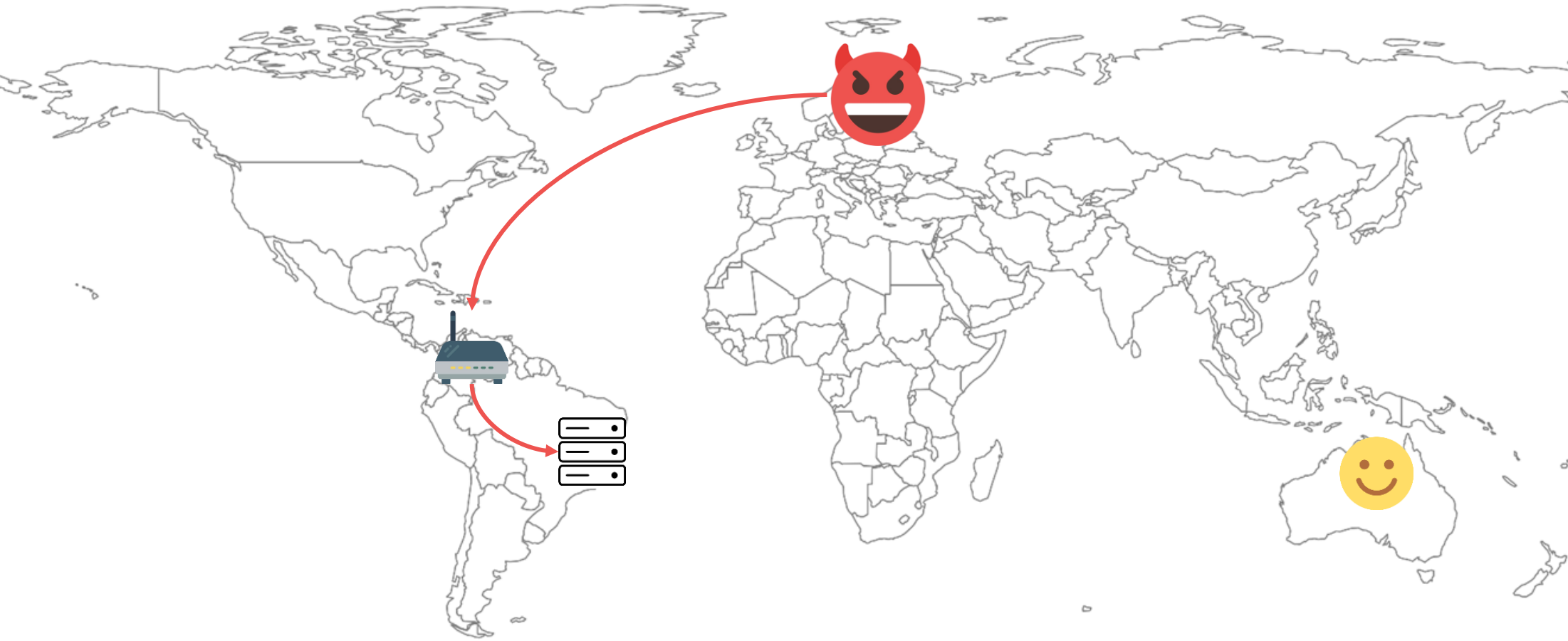
# Transparent Forwarders can be Utilized to Circumvent Rate Limiting

An attacker can orchestrate transparent forwarders



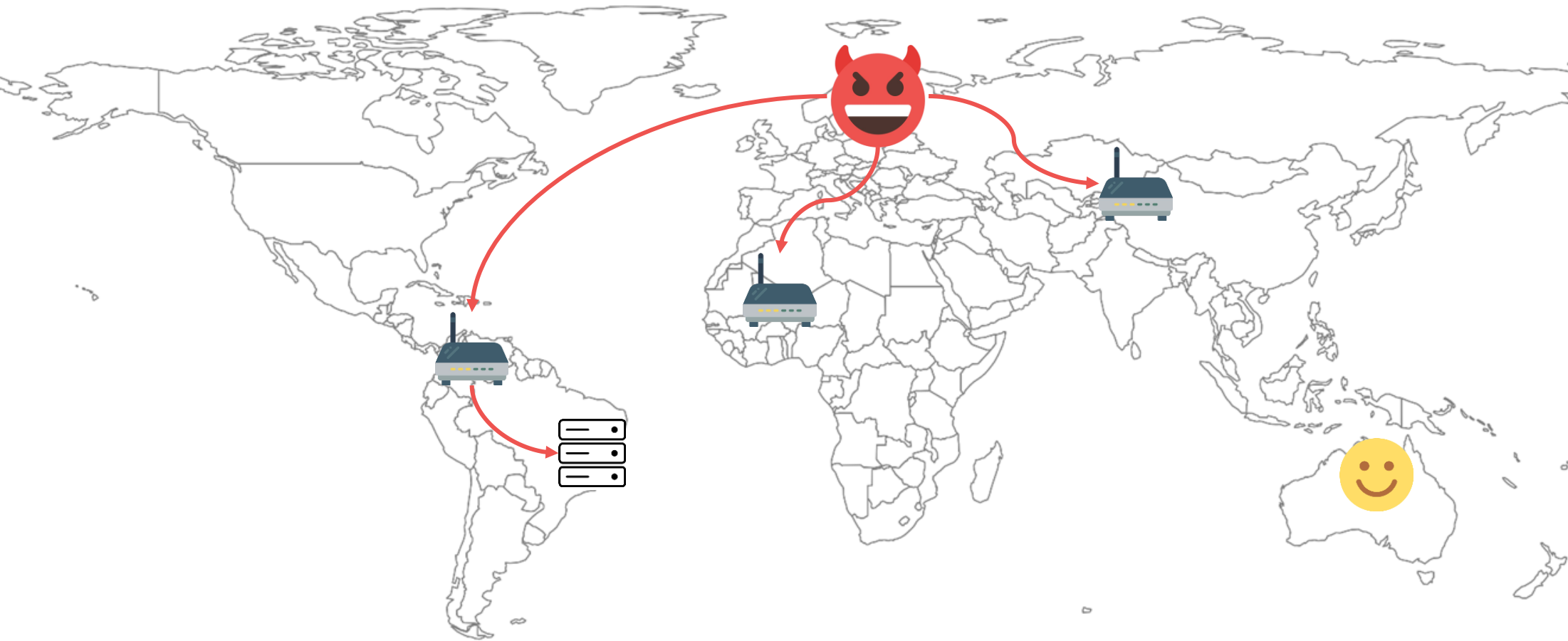
# Transparent Forwarders can be Utilized to Circumvent Rate Limiting

An attacker can orchestrate transparent forwarders to reach multiple PoP instances of public DNS providers such as Google.



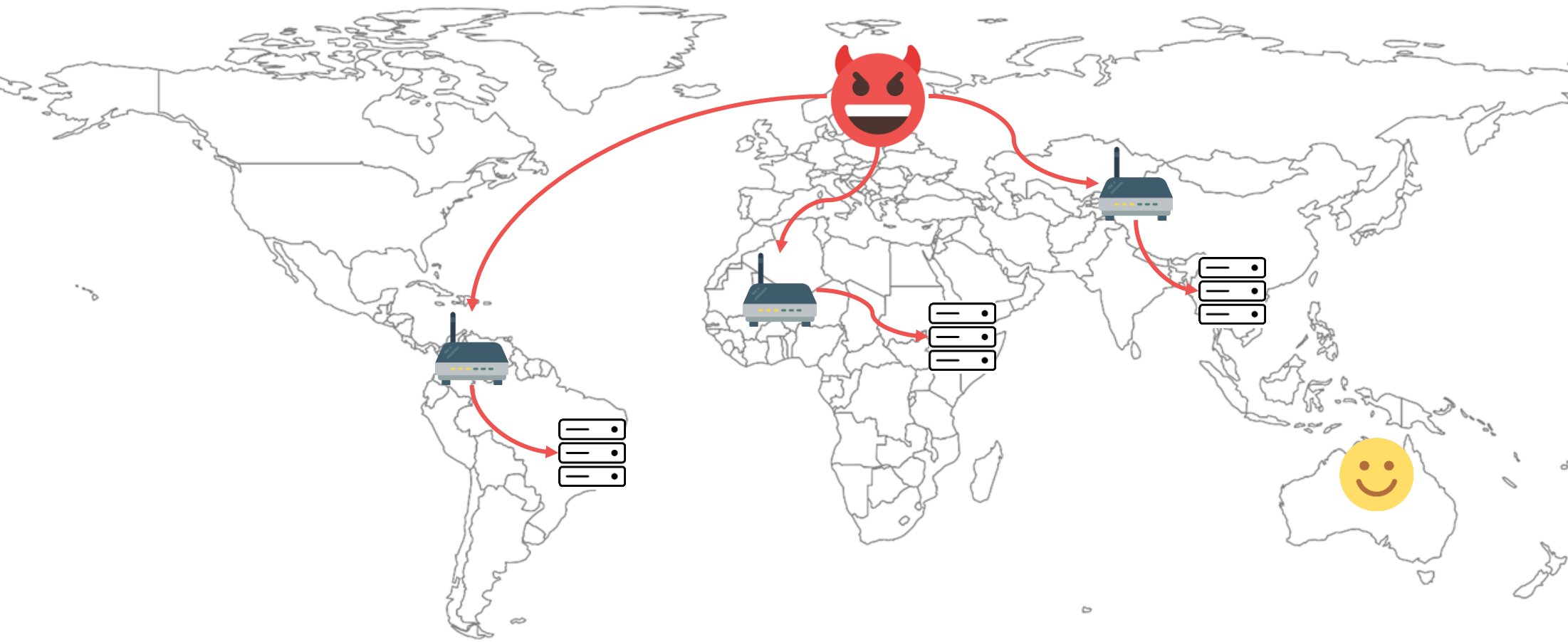
# Transparent Forwarders can be Utilized to Circumvent Rate Limiting

A set of transparent forwarders is selected



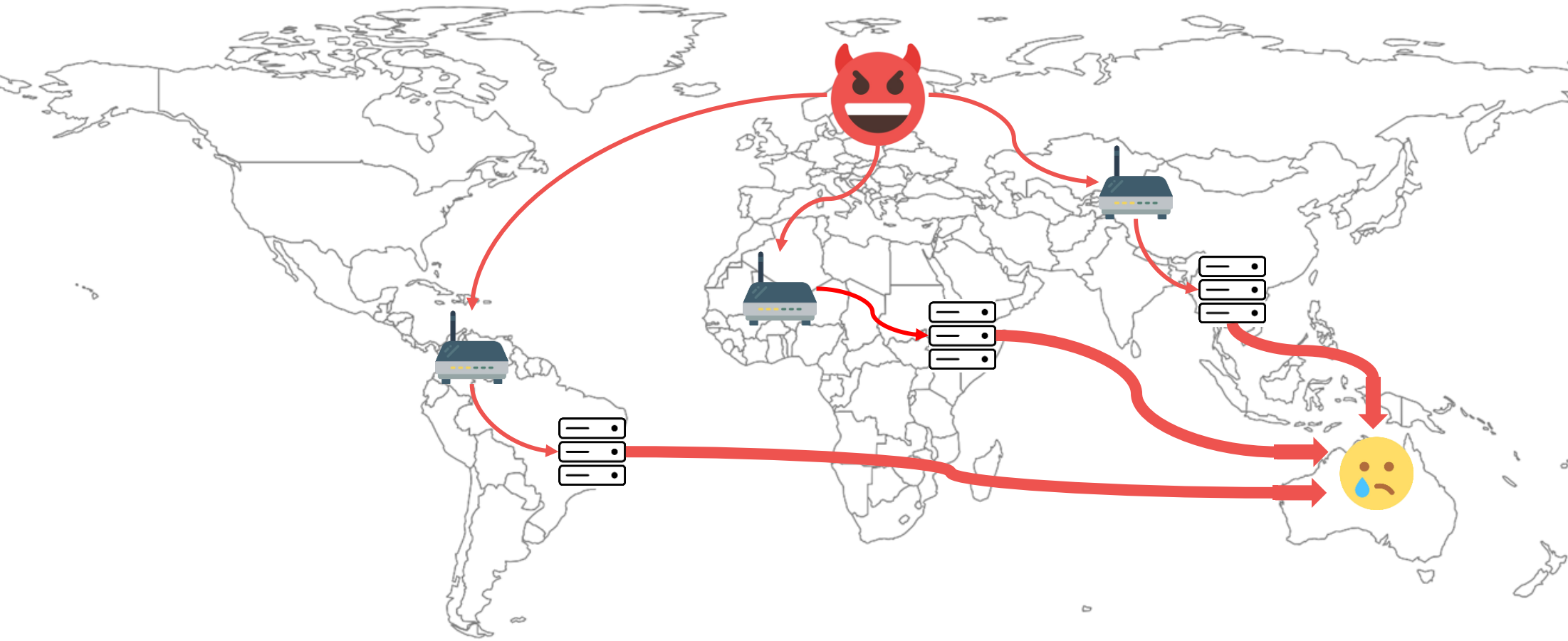
# Transparent Forwarders can be Utilized to Circumvent Rate Limiting

A set of transparent forwarders is selected that hits multiple PoPs.



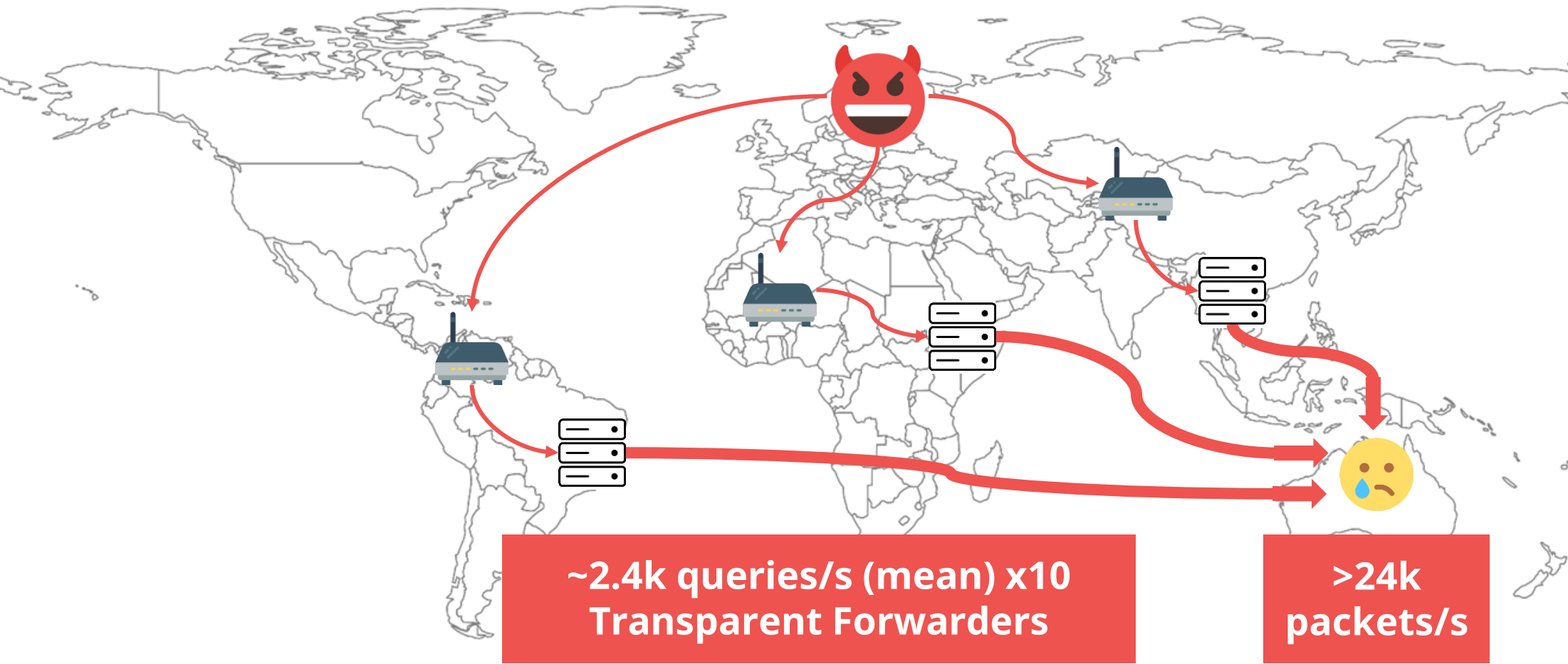
# Transparent Forwarders can be Utilized to Circumvent Rate Limiting

The victim receives a flood of DNS responses from, e.g., Google.



# Transparent Forwarders can be Utilized to Circumvent Rate Limiting

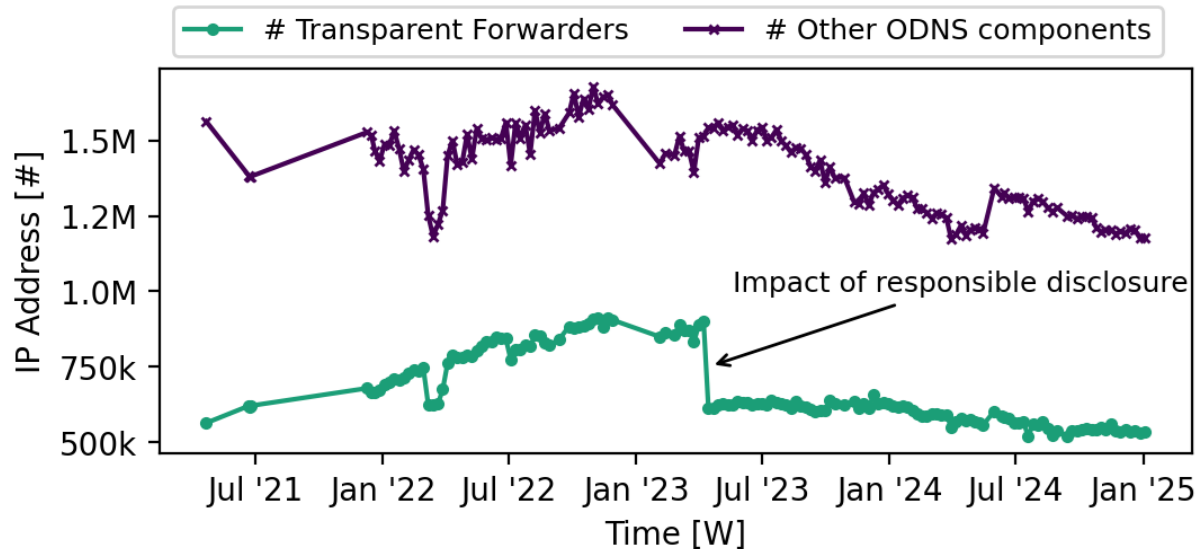
Using Internet-wide measurements, we can show an amplification factor of up to 14 by orchestrating 10 transparent forwarders.



# Agenda

1. Additional Attack Infrastructure
2. Higher Efficiency in Potential Attacks
3. Avoiding Local Rate Limits by Orchestrating Transparent Forwarders
4. **Responsible Disclosure and Mitigation Options**

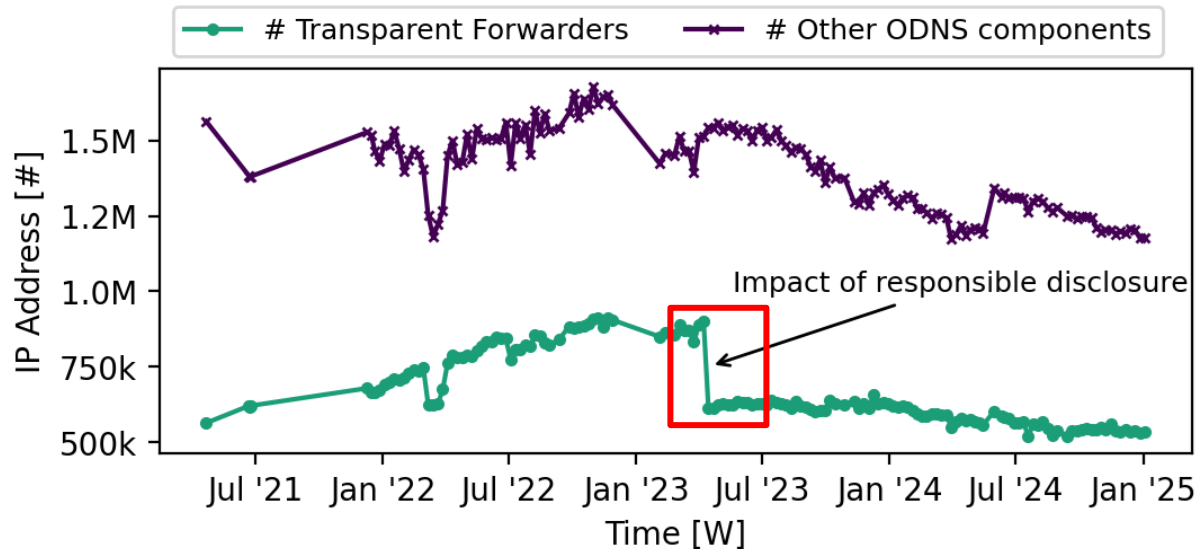
# Responsible Disclosure to Reduce the Impact of Transparent Forwarders



**Contacting operators.** We constantly try to get in touch with operators of networks that host transparent forwarders.

**Reducing the impact.** Our responsible disclosure removed >250k transparent forwarders from the threat landscape.

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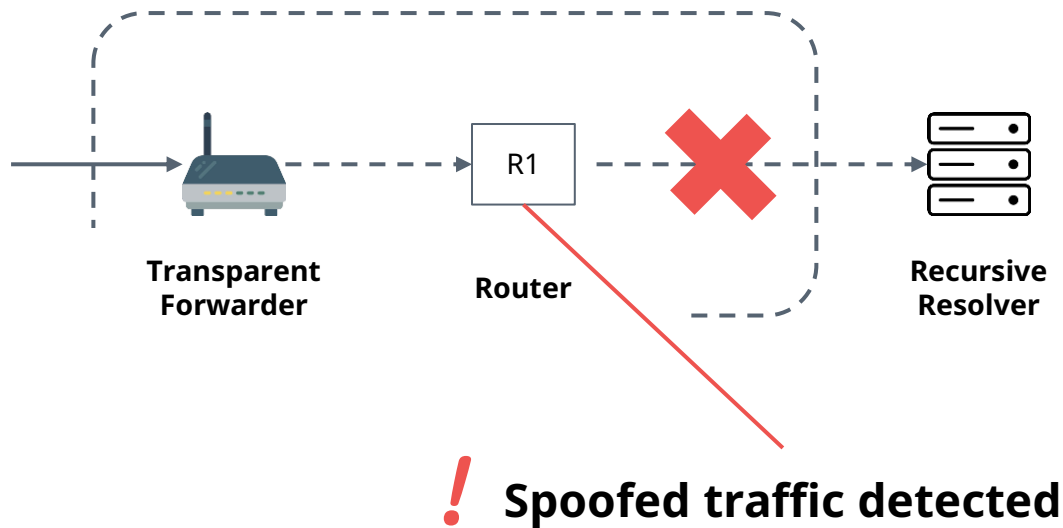
**Reducing the impact.** Our responsible disclosure removed >250k transparent forwarders from the threat landscape.

**A large European ISP fixed their firewall rules to prevent unrestricted access from transparent forwarders.**

**Operators should verify and update their firewall rules if necessary!**

# Mitigation by Network Ingress Filtering

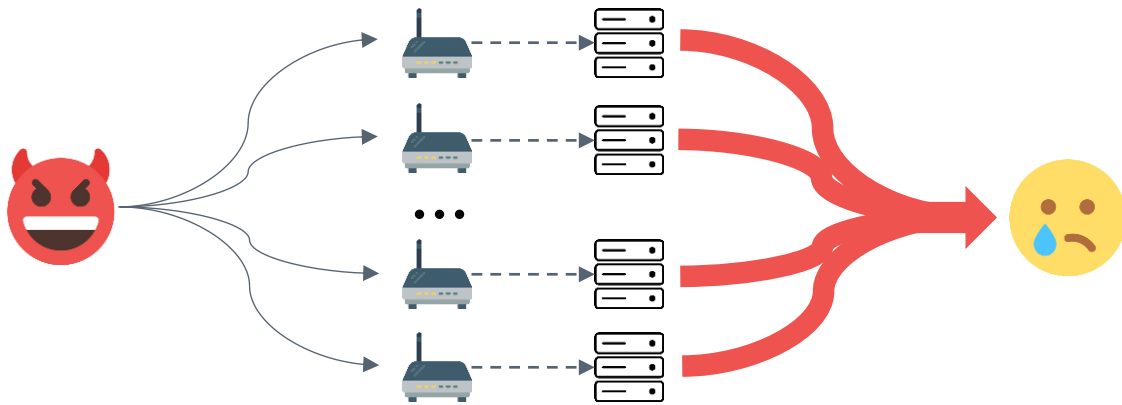
Network with transparent forwarder



**Networks with transparent forwarders** do not follow **BCP38** as they allow spoofed traffic to pass through the network.

**Solution.** Implement network ingress filtering; ensuring that only traffic from allowed ranges passes through.

# Mitigation by Strict Rate Limiting



**Local rate limiting.** Public DNS providers should strictly limit DNS replies, e.g., by limiting the number of queries per second.

**Global rate limiting.** Anycast public DNS providers should always implement a global monitoring strategy to detect anomalies.

# Conclusion

Transparent forwarders are a serious threat to the public Internet.

1. Transparent forwarders are **misconfigured devices**

**We update our dataset regularly and  
make it publicly available.**

<https://odns.secnw.net/data>



# Conclusion

Transparent forwarders are a serious threat to the public Internet.

1. Transparent forwarders are **misconfigured devices**
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2. Transparent forwarders allow for significantly **higher attack traffic**
3. Transparent forwarders can be orchestrated to **circumvent local rate limiting**
4. Transparent forwarders allow to **bypass shielded resolvers**
5. Eliminating the impact of transparent forwarders **requires little changes**

**We update our dataset regularly and  
make it publicly available.**

<https://odns.secnow.net/data>



# More details

## Transparent Forwarders: An Unnoticed Component of the Open DNS Infrastructure

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Freie Universität Berlin  
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### ABSTRACT

In this paper, we revisit the open DNS (ODNS) infrastructure and, for the first time, systematically measure and analyze transparent forwarders, DNS components that transparently relay between stub resolvers and recursive resolvers. Our key findings include four takeaways. First, transparent forwarders contribute 26% (563k) to the current ODNS infrastructure. Unfortunately, common periodic scanning campaigns such as Shadowserver do not capture transparent forwarders and thus underestimate the current threat potential of the ODNS. Second, we find an increased deployment of transparent forwarders in Asia and South America. In India alone, the ODNS consists of 80% transparent forwarders. Third, many transparent forwarders relay to a few selected public resolvers such as Google and Cloudflare, which confirms a consolidation trend of DNS stakeholders. Finally, we introduce DNSRoute++, a new traceroute approach to understand the network infrastructure connecting transparent forwarders and resolvers.

### CCS CONCEPTS

• **Networks** → **Public Internet; Security protocols; Network measurement; • Security and privacy** → *Security protocols.*

Table 1: Comparison of known open DNS components.

|                  | 2014      | 2020  | 2021  |      |      |            |
|------------------|-----------|-------|-------|------|------|------------|
|                  | [26]      | [1]   | [8]   | [39] | [38] | This Work  |
| # Rec. Resolvers | n/a       | 20K   | 50K   | n/a  | n/a  | 32K (2%)   |
| Forwarders       |           |       |       |      |      |            |
| # Recursive      | n/a       | 1.4M  | 1.7M  | n/a  | n/a  | 1.5M (72%) |
| # Transparent    | 0.6M (2%) | n/a   | n/a   | n/a  | n/a  | 0.6M (26%) |
| All ODNSes       | 25.6M     | 1.42M | 1.75M | 1.8M | 1.6M | 2.125M     |

located in a remote network. This “openness” makes the ODNS system a popular target for attackers, who are in search for amplifiers of DNS requests, for periodic DNS scan campaigns, which try to expose the attack surface, and for researchers, who want to learn more about DNS behavior.

Originally observed in 2013 [31], transparent DNS forwarders have not been analyzed in detail since then, but fell off the radar in favor of recursive forwarders and resolvers. This raises concerns for two reasons. First, the relative amount of transparent forwarders



## Forward to Hell? On the Potentials of Misusing Transparent DNS Forwarders in Reflective Amplification Attacks

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### Abstract

The DNS infrastructure is infamous for facilitating reflective amplification attacks. Various countermeasures such as server shielding, access control, rate limiting, and protocol restrictions have been implemented. Still, the threat remains throughout the deployment of DNS servers. In this paper, we report on and evaluate the often unnoticed threat that derives from transparent DNS forwarders, a widely deployed, incompletely functional set of DNS components. Transparent DNS forwarders transfer DNS requests without rebuilding packets with correct source addresses. As such, transparent forwarders feed DNS requests into (mainly powerful and anycasted) open recursive resolvers, which thereby can be misused to participate unwillingly in distributed reflective amplification attacks. We show how transparent forwarders raise severe threats to the Internet infrastructure. They easily circumvent rate limiting and achieve an additional, scalable impact via the DNS anycast infrastructure. We empirically verify this scaling behavior up to a factor of 14. Transparent forwarders can also assist in bypassing firewall rules that protect recursive resolvers, making these shielded infrastructure entities part of the global DNS attack surface.

### CCS Concepts

• **Networks** → **Public Internet; Network measurement; • Security and privacy** → **Denial-of-service attacks.**

### Keywords

DNS, Open DNS, ODNS, DDoS, DRDoS, Internet Measurements

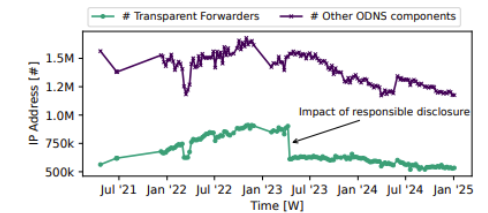


Figure 1: Number of transparent DNS forwarders and other open DNS components per week from June 2021 to January 2025. By sharing our results with network operators, we were able to decrease the impact of transparent forwarders by 30%.

### 1 Introduction

The open DNS (ODNS) infrastructure is a valuable target for attackers to misuse them in DNS reflective amplification attacks. While DNS service providers implement various countermeasures, the major part of the ODNS is unintentionally open to anyone, hence lacking of proper security configurations. Originally observed in 2013 [16] and measured in detail in 2021 [22], transparent DNS forwarders, which forward DNS requests to their configured resolver without rewriting the source IP address, currently account for 30% of the ODNS infrastructure, see Figure 1. Despite their

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[ACM CCS '25, <https://doi.org/10.1145/3719027.3765096>]